



RS6x_7x mmWave Radar Evaluation Quick Start Guide

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Revision History

Version	Description	Date
1.5	Added the RS6241 communication-mode configuration to Table 2-2; Corrected descriptions that omitted RS6241;	2026-8-13
1.4	Added the user-coordinate-system definition for the top-mounted development-board orientation; Added instructions for obtaining calibration values from Device Flash; Renamed the micro-motion point cloud to the presence point cloud; Corrected the IRQ DIP-switch description in UART-I ² C mode; Added RS6241 support; Added presence point-cloud performance configuration; Added descriptions of ADC and range-FFT data acquisition;	2026-6-11
1.3	Added detailed data-visualization descriptions; Added the DataCube storage format and usage/post-processing descriptions;	2025-12-17
1.2	Added CFAR algorithm-selection guidance; Added recommended waveform-configuration constraints to Section 3.1;	2025-10-21
1.1	Updated the user-coordinate-system definition and moved the device-coordinate-system description to development documentation; Updated the definitions of Side-Mounted Vertical and Side-Mounted Horizontal; Corrected the DUAL-UART DIP-switch description; Added instructions for reading DataCube with RadarAnalysisTool for data analysis;	2025-10-07
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1 Purpose and Scope

This document describes how to use the Possumic RS6x_7x Series mmWave Radar Evaluation Kit. By visualizing key radar data, developers can quickly determine whether an RS6x_7x Series radar meets application requirements.

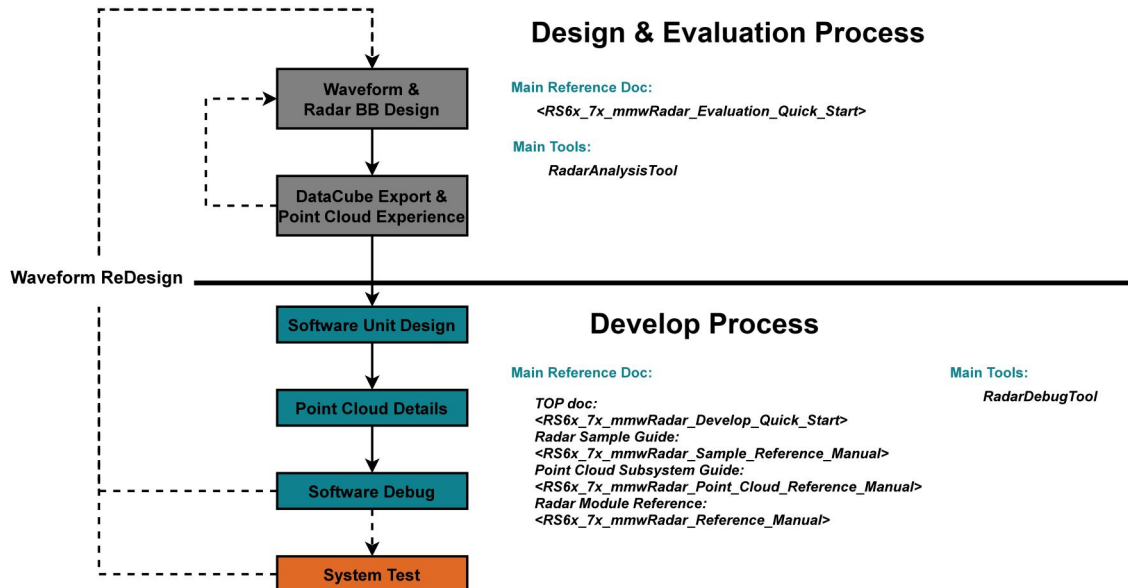


Figure 1-1 Recommended Radar Evaluation, Design, and Development Workflow and Tools

Developers are advised to use the SDK as shown in Figure 1-1. The SDK divides the workflow into design and evaluation and development phases, and provides corresponding tools.

During evaluation, RadarAnalysisTool and its firmware are used to define radar waveforms, acquire waveforms or point clouds on the Host, and verify and optimize custom algorithms. RadarAnalysisTool cannot fully configure point-cloud processing during evaluation, so its point-cloud output is for basic evaluation only. For complete point-cloud performance, use the Tracking, Posture, or Sleep firmware with RadarDebugTool.

During development, use SDK APIs and tools and refer to the Samples and SDK documentation to deploy the selected waveform and custom algorithms to the embedded system for testing, verification, and optimization.

Supported devices: RS6130, RS6240, and RS6241.

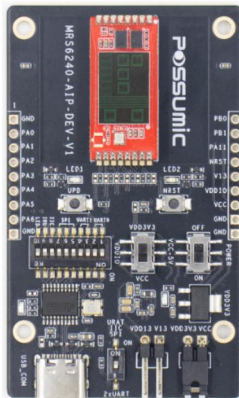
2 Setting Up the Hardware Environment

RadarAnalysisTool requires a matching module and development board. Device, module, and board combinations are shown in Table 2-1 and Figure 2-1.

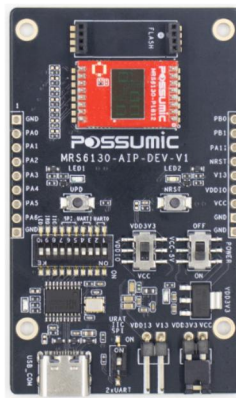
Table 2-1 Module and Development-Board Mapping

Device	Module	Development Board
RS6130	MRS6130-P1812	MRS6130-AIP-DEV-V1
RS6130	MRS6130-P1806	MRS6130-AIP-DEV-V1
RS6240	MRS6240-P2512	MRS6240-AIP-DEV-V1
RS6241	MRS6241-P2828-M62	MRSx241-DEV-V1
RS6241	MRS6241-P2840-M81	MRSx241-DEV-V1

CHIP: RS6240
MODULE: MRS6240-P2512
DEV BOARD: MRS6240-AIP-DEV-V1



CHIP: RS6130
MODULE: MRS6130-P1812
DEV BOARD: MRS6130-AIP-DEV-V1



CHIP: RS6130
MODULE: MRS6130-P1806
DEV BOARD: MRS6130-AIP-DEV-V1



CHIP: RS6241
MODULE: MRS6241-P2828-M62
DEV BOARD: MRSx241-DEV-V1



CHIP: RS6241
MODULE: MRS6241-P2840-M81
DEV BOARD: MRSx241-DEV-V1

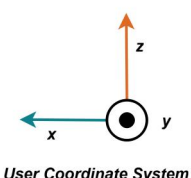
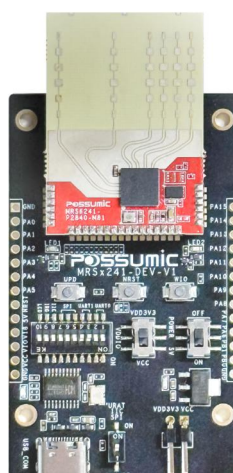


Figure 2-1 Modules and Development Boards

2.1 Installing the Driver and Preparing the Tool

Before proceeding, ensure that the UART driver is installed correctly.

- UART driver path: [/Software Kit/03 Tool/USB Serial Driver](#)

Install the UART driver. Depending on the DIP-switch settings, Device Manager displays two UART ports in DUAL-UART mode or one UART port in UART-SPI mode.



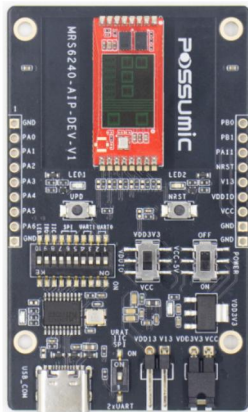
Figure 2-2 Verifying the UART Driver

- RadarAnalysisTool
RadarAnalysisTool is the primary evaluation tool. Path: [/Software Kit/03 Tool/RadarAnalysisTool](#)
- Burn the RadarAnalysisTool firmware to the device to exchange data with RadarAnalysisTool. Firmware path: [/Software Kit/04 Image/radar analysis \[board name\].img](#)

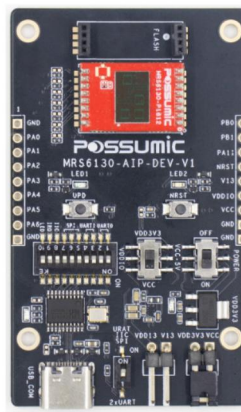
2.2 Development-Board Orientation Definitions

RadarAnalysisTool supports side-mounted vertical, side-mounted horizontal, and top-mounted orientations. The user-coordinate origin is at the device center. For the side-mounted orientations in Figures 2-3 and 2-4, when viewed from behind the radar toward the detection direction, right is +x, forward is +y, and up is +z. For the top-mounted orientation in Figure 2-5, when viewed from the USB port toward the radar module, right is +x, forward is +y, and up is +z. All RadarAnalysisTool point-cloud outputs are converted to the user coordinate system.

CHIP: RS6240
MODULE: MRS6240-P2512
DEV BOARD: MRS6240-AIP-DEV-V1



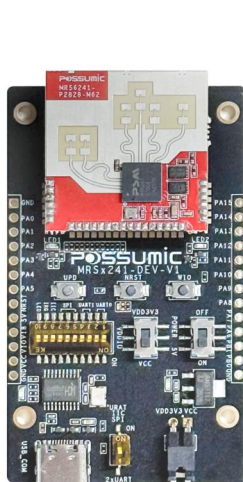
CHIP: RS6130
MODULE: MRS6130-P1812
DEV BOARD: MRS6130-AIP-DEV-V1



CHIP: RS6130
MODULE: MRS6130-P1806
DEV BOARD: MRS6130-AIP-DEV-V1



CHIP: RS6241
MODULE: MRS6241-P2828-M62
DEV BOARD: MRSx241-DEV-V1



CHIP: RS6241
MODULE: MRS6241-P2840-M81
DEV BOARD: MRSx241-DEV-V1

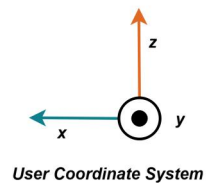
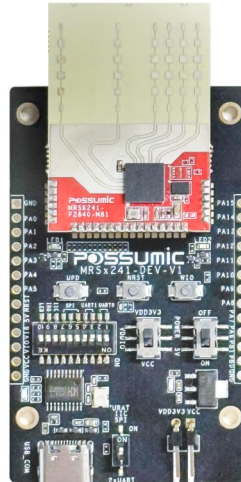
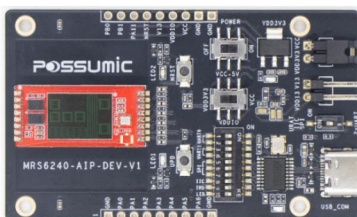
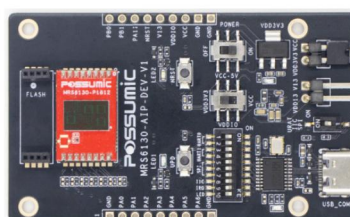


Figure 2-3 Side-Mounted Vertical Development-Board Orientation

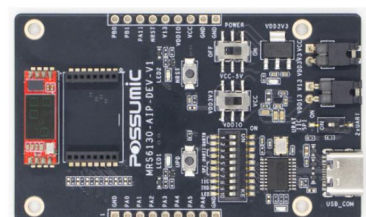
CHIP: RS6240
MODULE: MRS6240-P2512
DEV BOARD: MRS6240-AIP-DEV-V1



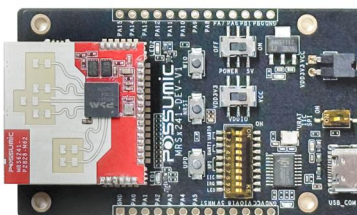
CHIP: RS6130
MODULE: MRS6130-P1812
DEV BOARD: MRS6130-AIP-DEV-V1



CHIP: RS6130
MODULE: MRS6130-P1806
DEV BOARD: MRS6130-AIP-DEV-V1



CHIP: RS6241
MODULE: MRS6241-P2828-M62
DEV BOARD: MRSx241-DEV-V1



CHIP: RS6241
MODULE: MRS6241-P2840-M81
DEV BOARD: MRSx241-DEV-V1

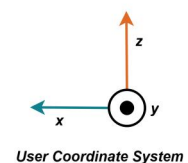
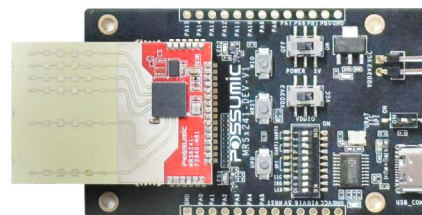


Figure 2-4 Side-Mounted Horizontal Development-Board Orientation

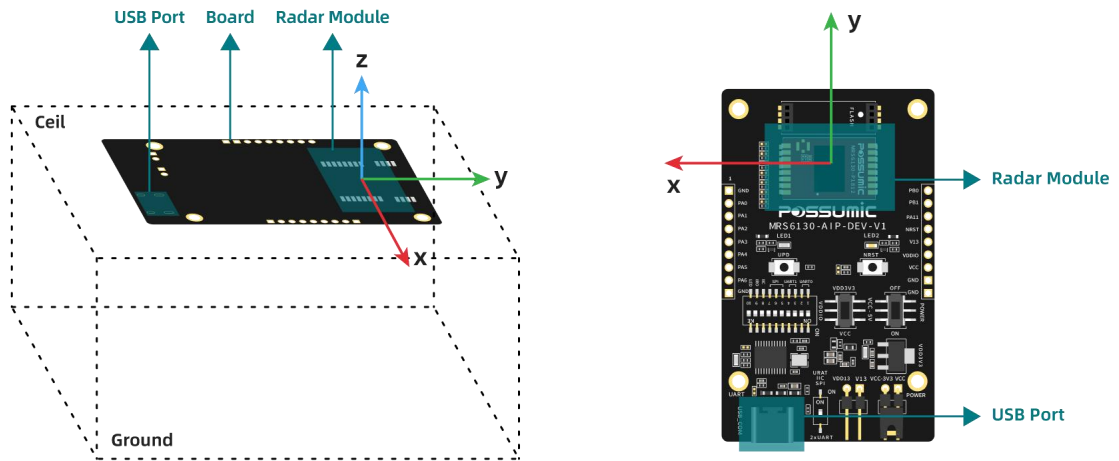


Figure 2-5 Top-Mounted Development-Board Orientation

Tip 1: For accurate evaluation, ensure that the radar orientation used during design and evaluation is consistent with the device orientation used during development. Otherwise, development performance may not match the evaluation results.

2.3 Configuring the Development Board

Before connecting the development board to the PC, verify that it is configured for the correct operating mode.

Each board has two functionally identical DIP-switch groups, indicated by orange and green boxes. The orange group is FUNC SEL and the green group is MODE SEL. Also note the POWER single-pole double-throw switch and the UPD (UPDATE) tactile switch.

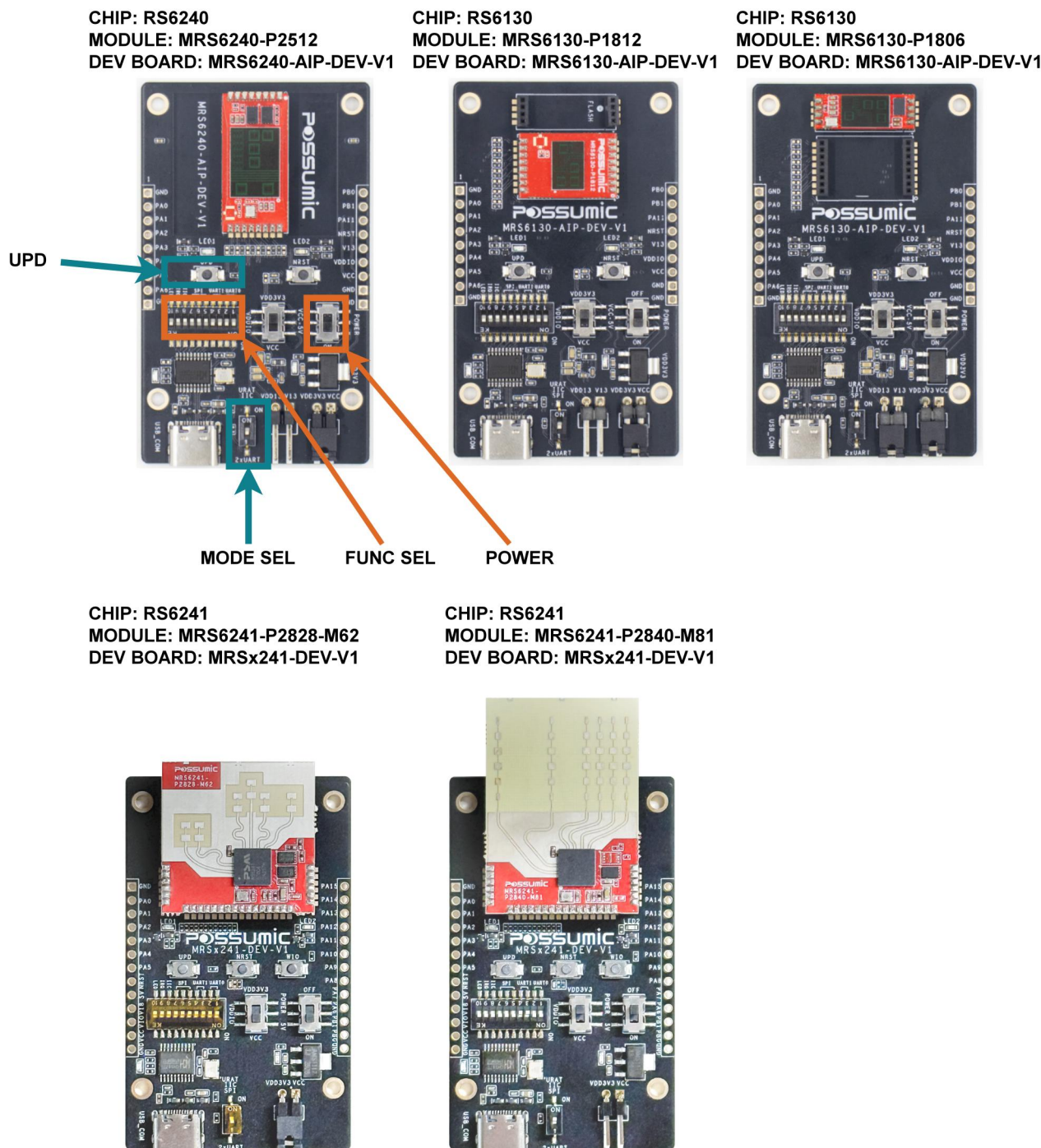


Figure 2-6 Module DIP Switches and Buttons

The MRS6240-AIP-DEV-V1 and MRS6130-AIP-DEV-V1 boards support DUAL-UART, UART-SPI, and UART-I²C modes, controlled by FUNC SEL and MODE SEL.

Table 2-2 Communication-Mode Functions for Different Modules

Mode Selection	DUAL-UART			UART-SPI/UART-IIC		
Hardware Interface	UART0	UART1	SPI	UART0	UART1	SPI/IIC
MRS6130-P1806	Tx/Rx	Tx ONLY	-	Tx/Rx	-	NOT SUPPORT
MRS6130-P1812	Tx/Rx	Tx/Rx	-	Tx/Rx	-	Tx/Rx
MRS6240-P2512	Tx/Rx	Tx/Rx	-	Tx/Rx	-	Tx/Rx

MRS6241-P2828	Tx/Rx	Tx/Rx	-	Tx/Rx	-	Tx/Rx
MRS6241-P2840	Tx/Rx	Tx/Rx	-	Tx/Rx	-	Tx/Rx

The DIP-switch positions for each mode are listed below:

Table 2-3 DIP-Switch Settings for Different Modes

Switch Position	FUNC SEL										MODE SEL
	1	2	3	4	5	6	7	8	9	10	1
	UART 0	UART 0	UART 1	UART 1	SPI	SPI	SPI	IIC	IRQ	LED	2xUART
DUAL-UART	ON	ON	ON	ON	OFF	OFF	OFF	OFF	OFF	ANY	OFF
UART-SPI	ON	ON	OFF	OFF	ON	ON	ON	OFF	ON	ANY	ON
UART-IIC	ON	ON	OFF	OFF	OFF	OFF	OFF	ON	ON	ANY	ON

ON means that the DIP switch is set toward the position marked ON.

SPI supports up to 60 Mbps depending on CPU frequency, allowing a 256 KB DataCube at approximately 10 Hz. UART supports up to 2 Mbps. If bit errors occur because 2 Mbps is unstable on some PCs, use 1 Mbps.

During evaluation, configure MRS6130-P1812 and MRS624x modules for UART-SPI mode and MRS6130-P1806 for DUAL-UART mode.

After changing MODE SEL, power-cycle the module for the new mode to take effect.

For more hardware information, see the user manual for the corresponding development board.

2.4 Downloading Firmware

Download the RadarAnalysisTool firmware to the board from [/Software Kit/04 Image/radar_analysis.img](#).

1. Connect the board to the PC through USB and set POWER to OFF.
2. Hold UPD while setting POWER to ON, and continue holding UPD for at least 2 seconds. The radar enters Bootloader mode.
3. Open RadarAnalysisTool, select the platform, select the control COM port (the higher-numbered port for MRS6130-P1806 in DUAL-UART mode), set the baud rate to 115200, select the corresponding RadarAnalysisTool firmware, and click Connect and Download. For troubleshooting, see the ['RS6x_7x_RadarAnalysisTool_User_Manual'](#).

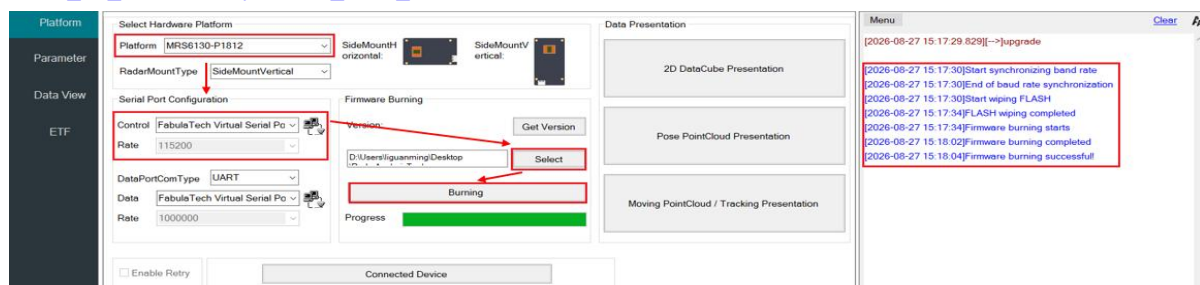


Figure 2-7 Firmware Download

After the burn is complete, power-cycle the radar to run the RadarAnalysisTool firmware.

2.5 Verifying Firmware Operation

If the firmware was downloaded correctly, configure the signal to display in RadarAnalysisTool and walk in front of the radar. Point clouds generated by the moving person should be visible. The following procedure uses the MRS6240-P2512 after the download shown in Figure 2-7.

1. Orient the board side-mounted vertically as shown in Figure 2-3 and configure UART-SPI mode according to Table 2-3. Configure MRS6130-P1806 for DUAL-UART mode.

- Place the radar on a stable base and do not touch the USB cable, which may cause packet loss or connection errors.
- Select the hardware platform, Side-Mounted Vertical orientation, and control port. Select SPI for the data interface (UART at 1000000 baud is recommended for MRS6130-P1806), and click Connect Device. Device Connected Successfully is displayed. If the Device does not respond, wait several seconds for initialization and reconnect. For persistent failures, see the '[RS6x_7x_RadarAnalysisTool_User_Manual](#)'.

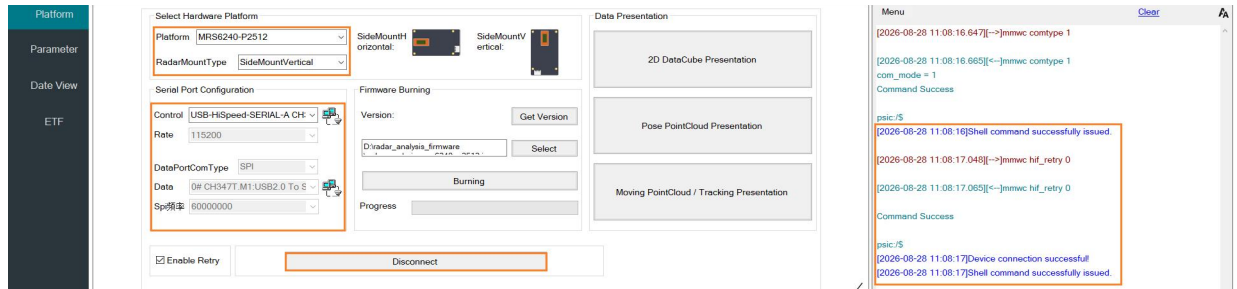


Figure 2-8 Configuring RadarAnalysisTool for SPI Mode

- On Platform Configuration, click Motion Point Cloud/Tracking Display. RadarAnalysisTool configures the radar and switches to Data View. Walk in front of the radar or move the radar to observe point-cloud data.

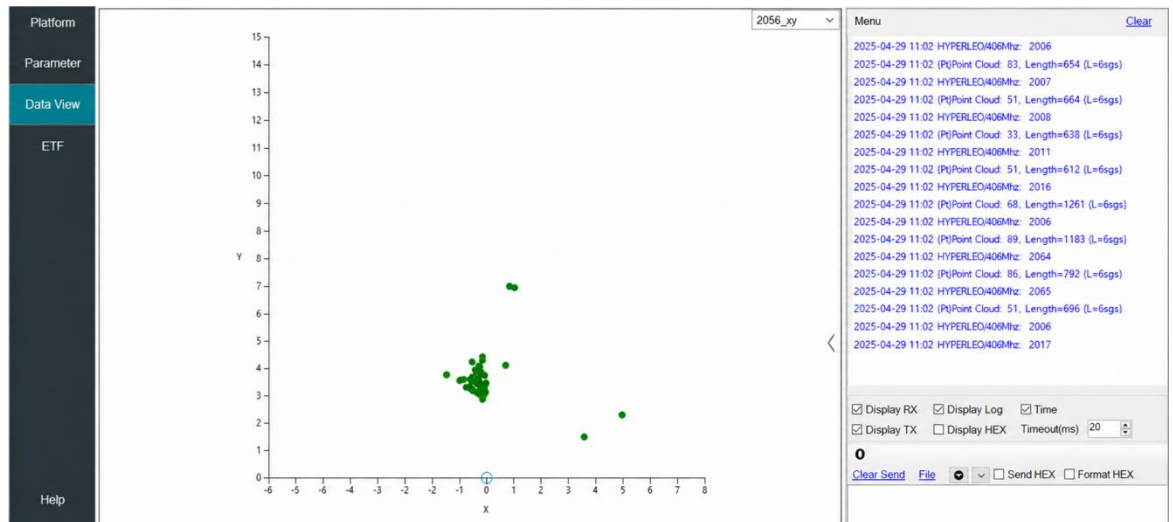


Figure 2-9 Pedestrian Point-Cloud Data

3 Evaluation Using RadarAnalysisTool

This chapter describes how to evaluate performance and the dimensions to assess. For tool operations, see the '[RS6x_7x_RadarAnalysisTool_User_Manual](#)'.

Before evaluation, complete the hardware setup in Chapter 2 and download the RadarAnalysisTool firmware.

The RadarAnalysisTool firmware data flow is shown below:

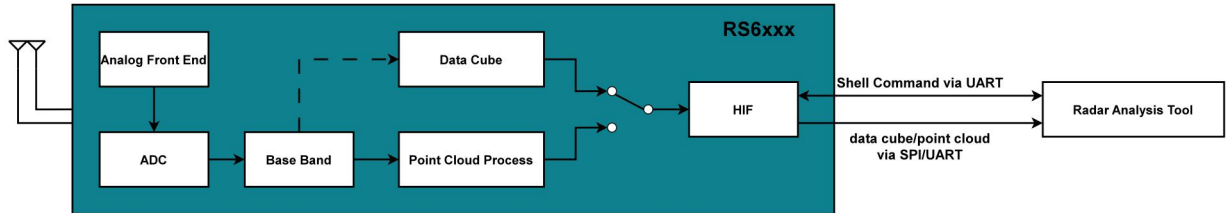


Figure 3-1 RadarAnalysisTool Data Flow

RadarAnalysisTool provides two primary outputs: DataCube and point cloud. During design and evaluation, DataCube is generally configured as the output of the 2D-FFT flow: a 3D IQ matrix of range FFT points * Doppler FFT points * transmit antennas * receive antennas. For point-cloud output, the firmware executes the point-cloud processing algorithm and reports point-cloud information for basic evaluation.

3.1 Configuring Range and Velocity Resolution

On Parameter Configuration, configure maximum range, range resolution, maximum velocity, velocity resolution, and related parameters.

Figure 3-2 Resolution Configuration

An evaluation waveform should maximize sensitivity to weak signals so that small abnormal signals can be identified. Therefore, configure as many range and Doppler FFT points as practical.

The range FFT point count is calculated as follows:

$$2^{\text{ceil}(\log 2 (\text{maximum range}/\text{range resolution}))}$$

The Doppler FFT point count is calculated as follows:

$$2^{\text{ceil}(\log 2 (2 * (\text{maximum velocity}/\text{velocity resolution}))}$$

During evaluation, $\log_2(\text{maximum range} / \text{range resolution})$ and $\log_2(2 * \text{maximum velocity} / \text{velocity resolution})$ should be integers; otherwise, spectral leakage may affect the results.

Avoid configuring a maximum detection range below 5 m or exactly equal to the required detection range. Prefer the following constraint:

Maximum detection range * 0.8 > required detection range, and maximum detection range > 5 m

Keep maximum range/range resolution > 64, corresponding to at least 128 range FFT points, for higher processing gain.

3.2 Memory Limits for Waveform Configuration

DataCube memory is limited. If the required storage exceeds the hardware limit, compression is applied, which may reduce dynamic range and distort the DataCube. **Avoid waveform configurations that trigger compression during evaluation.**

RS624x devices provide 256 KB of DataCube storage and RS6130 devices provide 128 KB. Each IQ value occupies 4 bytes (2-byte real and 2-byte imaginary). For M_TX transmitters, N_RX receivers, N_RFFT range FFT points, and N_DFFT Doppler FFT points, DataCube memory usage is:

$$M_TX \times N_RX \times N_RFFT \times N_DFFT \times 4 (\text{Bytes})$$

The factors affecting DataCube storage are the numbers of transmit and receive antennas and the range and Doppler FFT point counts.

During design and evaluation, use all transmit and receive antennas: 2T4R for RS624x and 1T3R for RS6130. Typical configurations that fill DataCube memory are:

Table 3-1 Typical Memory-Usage Configurations

Device	Transmit/Receive Antenna Selection	Range FFT Points	Doppler FFT Points	Memory Usage
RS624x	2T4R	256/128/64	32/64/128	$2*4*256*32*4=256K$
RS6130	1T3R	256/128/64	32/64/128	$1*3*256*32*4=96K$

RadarAnalysisTool displays a warning when a waveform exceeds 256 KB for RS624x or 128 KB for RS6130.

If more 2D-FFT points are required, use at most 2* compression. No additional software configuration is required; firmware enables it automatically when the waveform exceeds DataCube capacity. One IQ value is stored as 2 bytes in hardware and automatically decompressed when read, so software always receives the standard IQ type.

Table 3-2 Typical Memory-Usage Configurations (2* Compression)

Device	Transmit/Receive Antenna Selection	Range FFT Points	Doppler FFT Points	Memory Usage
RS624x	2T4R	256/128/64	64/128/256	$2*4*256*64*2=256K$
RS6130	1T3R	256/128/64	64/128/256	$1*3*256*64*2=96K$

2* compression reduces radar dynamic range and may raise quantization noise in a range bin containing a strong signal.

3.3 Importing and Exporting Data

Before starting radar transmission, select Data View > Auto Save > Menu/Data Save/Auto Save FFT Table. After transmission begins, RadarAnalysisTool automatically saves DataCube or point-cloud data as binary and CSV files for offline analysis. For parsing, see Data Format in the '[RS6x_7x_RadarAnalysisTool_User_Manual](https://www.possuimic.com/)'.

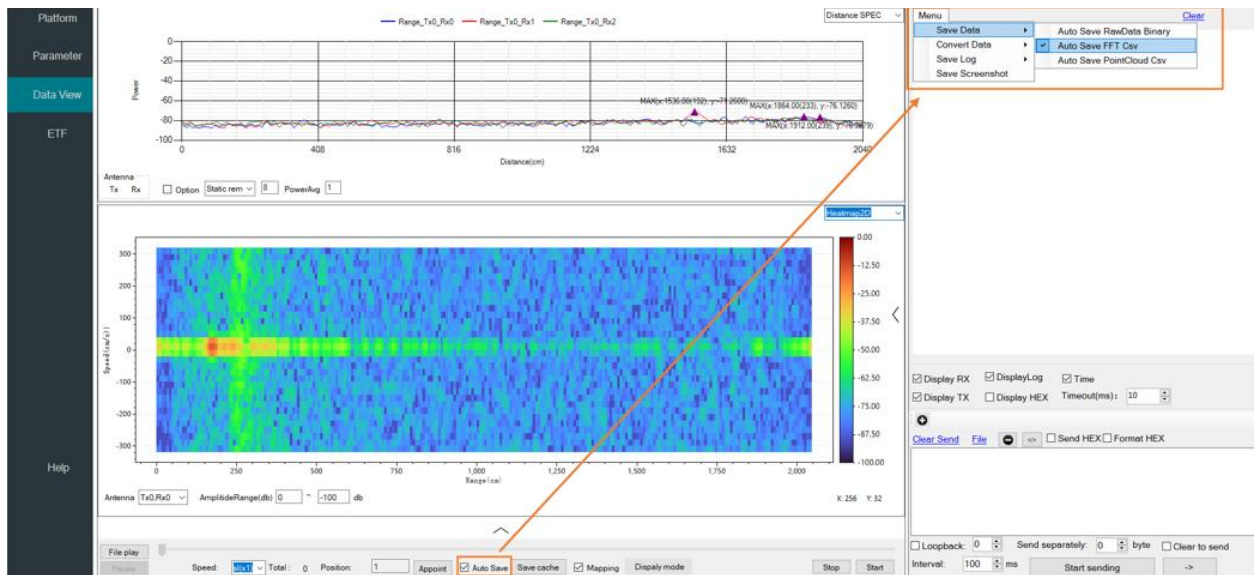


Figure 3-3 Saving Data with RadarAnalysisTool

3.4 Acquiring ADC and Range-FFT Data

RadarAnalysisTool can acquire ADC data. Disable RANGE FFT EN and DOP FFT EN so the radar skips range and Doppler FFT processing. The ADC data is reversed in range; apply `fft_shift` after the range FFT to obtain the correct range spectrum. For ADC post-processing, see the ['RS6x_7x_mmWave_Radar_Point_Cloud_Process_Reference_Manual'](#).

For ADC acquisition, maximum range must be a power-of-two integer multiple of range resolution, and maximum velocity must be a power-of-two integer multiple of velocity resolution.

Frame Structure Configuration					
Frame Structure:	2DFFT	Interval Num:	32	Frame Num:	0
Interval Chirp N:	1	Interval Period:	397 us	Frame Period:	1193 ms
				RANGE FFT EN	ON
				DOP FFT EN	ON

Figure 3-4 Configuring ADC Data Acquisition

If only DOP FFT EN is disabled, only the range FFT is performed, and maximum range must be a power-of-two integer multiple of range resolution.

When maximum velocity equals velocity resolution, the Doppler FFT point count is 2 and only DOP FFT EN can be disabled.

For detailed execution logic, see the RS6x_7x mmWave Radar Reference Manual.

4 Radar Performance Evaluation Methods

Radar performance evaluation focuses on the RF front end through baseband processing. It is essential when developers intend to implement their own radar algorithms with the SDK instead of using the reported point cloud.

After product hardware and mechanical assembly, use this chapter to benchmark against the development board and evaluate performance degradation on target hardware. For implementation, see the RS6x_7x mmWave Radar Development Quick Start Guide.

This chapter assumes familiarity with basic FMCW mmWave radar ranging and angle-estimation principles.

Key evaluation factors include:

1. Noise characteristics of all receive channels;
2. Amplitude and phase characteristics of target echoes on all receive channels.

Use RadarAnalysisTool visualization to identify abnormalities qualitatively and DataCube storage for further diagnosis and metric calculation, providing data for subsequent digital signal processing algorithm design.

If the waveform is not yet defined, after completing Chapter 2 connect the board and click 2D DataCube Display on Platform Configuration to configure the radar automatically and switch to Data View.

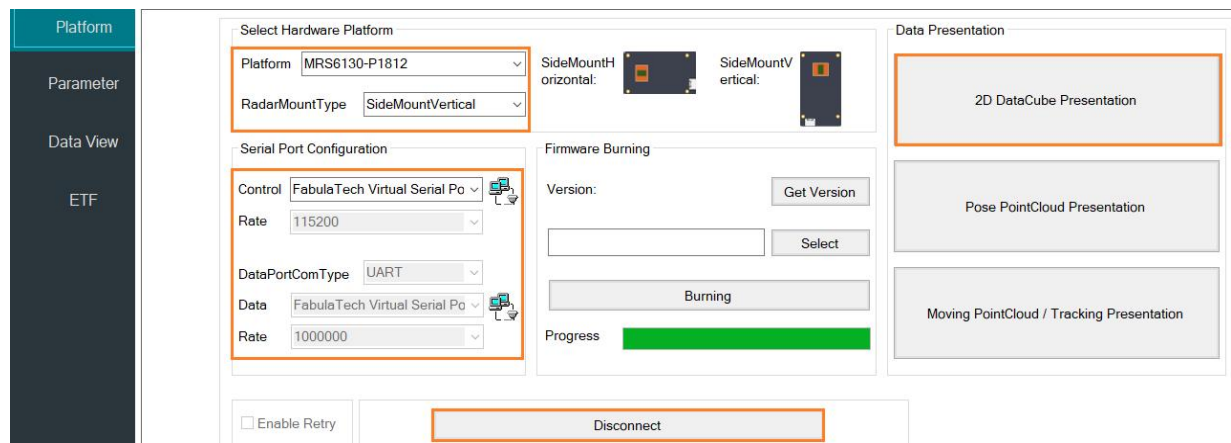


Figure 4-1 Quick Start for Noise Floor Performance Evaluation

4.1 Data Visualization Methods

Assuming that the number of range FFT points is N_{RFFT} and the number of Doppler FFT points is N_{DFFT} , RadarAnalysisTool supports the following evaluation methods:

- Range spectrum: For the 2D-FFT process, the "Range Spectrum" function performs simple processing on the DataCube and displays a plot with distance on the horizontal axis and power (dB) on the vertical axis. Range-spectrum analysis mainly uses 3 secondary processing methods: "Static Removal," "Single Frame," and "Smoothing."
- Static Removal: Calculate the power of each range-Doppler bin in the DataCube, remove the Doppler bins in $[N_{\text{DFFT}}/2-k, N_{\text{DFFT}}/2+k]$, where k is a configurable parameter, and then average the power along the Doppler dimension. This calculation is performed for each receive channel. Its purpose is to remove the effects of stationary targets and their leaked energy, allowing evaluation of noise-floor performance without target-energy interference or measurement of moving-target energy. The calculation process is as follows:

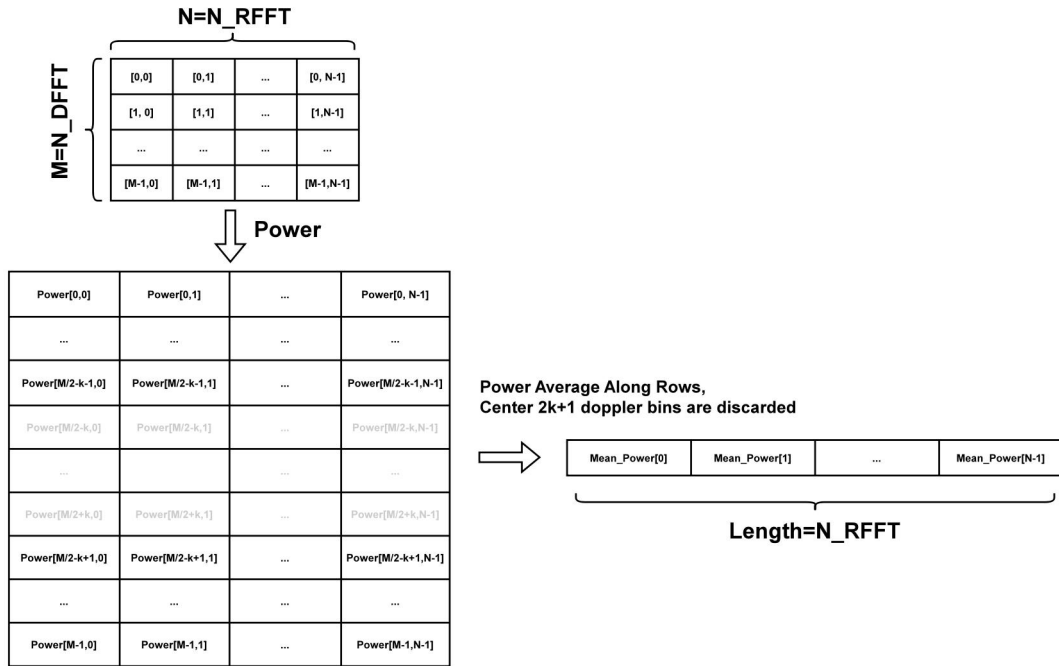


Figure 4-2 Static Removal Calculation Method

The following scenario illustrates how to observe the position and energy of a moving target: Select "Static Removal" for the range spectrum in "Data View." As a person moves away from the radar, a distinct peak corresponding to the person appears in the range spectrum. The speed-range heatmap shows a positive velocity, indicating that the person is moving away from the radar.

x:272(34):
-----range bin index = 34;
-----Distance of Human = 272cm(range bin index * RangeResolution)
y: -33.2430:
----- Normalized Power of Corner Reflector is -33.2430dBm

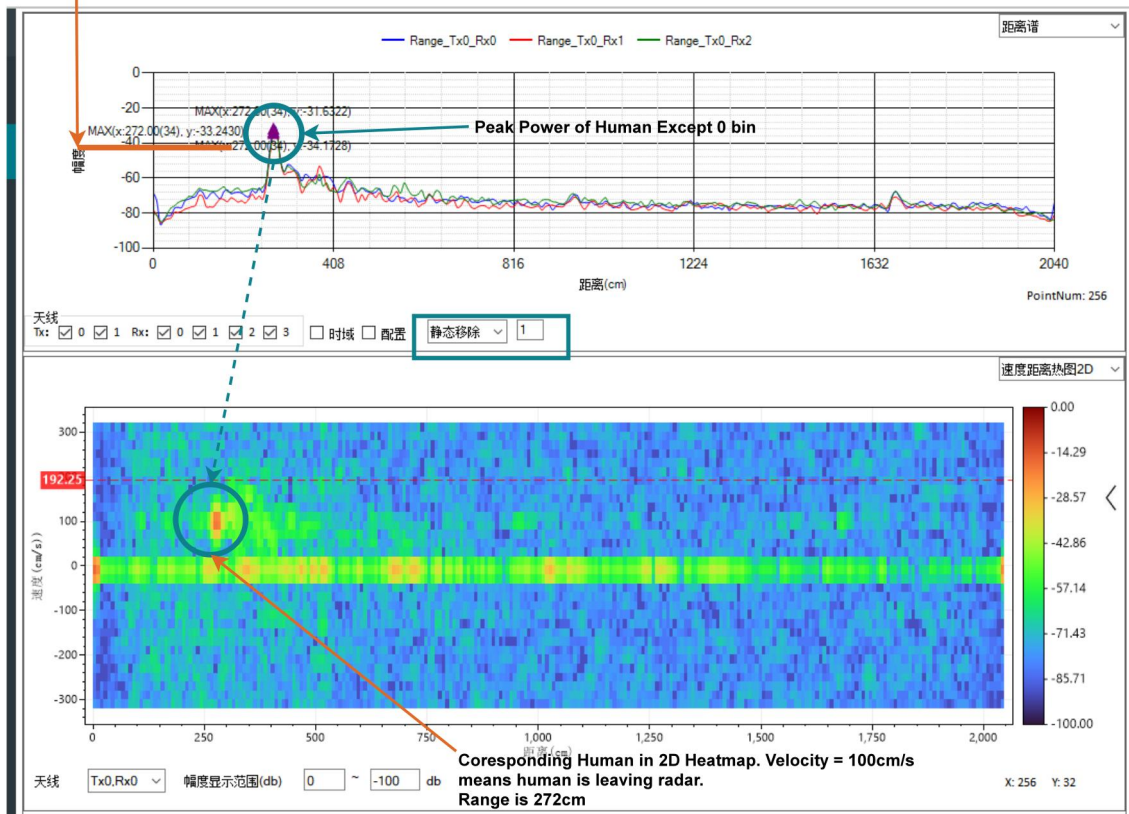


Figure 4-3 Observing a Moving Target Using Static Removal

- Single Frame: Used to observe the range spectrum of each Frame, with distance on the horizontal axis and power on the vertical axis. The power of all range bins at a velocity of 0 is calculated and

plotted. This range spectrum mainly shows the energy of targets that are stationary relative to the radar.

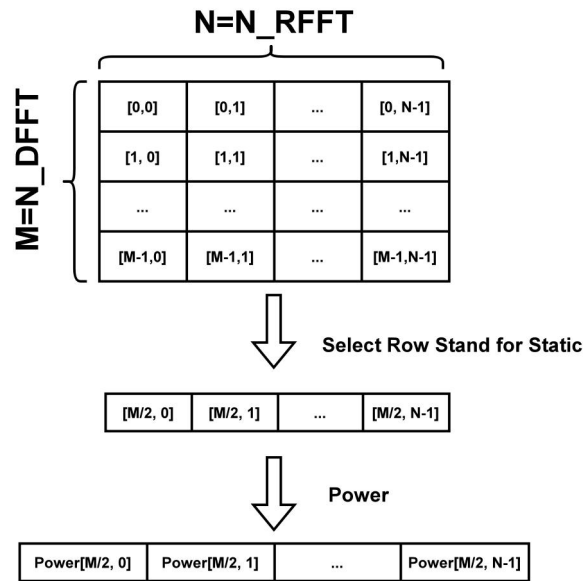


Figure 4-4 Single-Frame Calculation Method

- **Data Smoothing:** Based on the "Single Frame" result, calculates the average of the "Single Frame" results from the latest k Frames, where k is configurable. This method can be used to observe the energy of stationary targets more steadily.

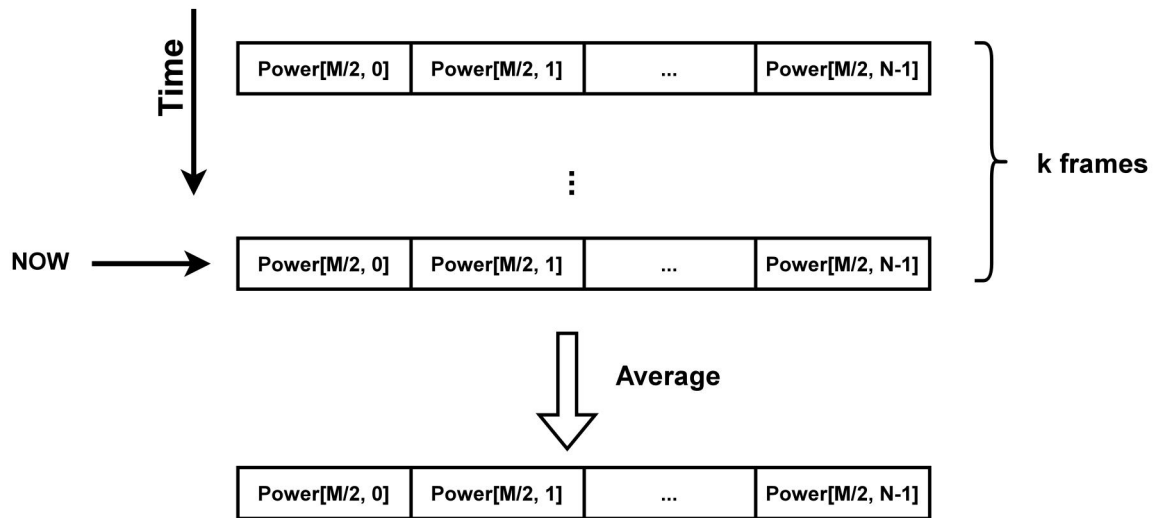


Figure 4-5 Data Smoothing Calculation Method

The following scenario illustrates how to observe target position and energy: Place a corner reflector 1.28 m in front of the radar. A distinct peak corresponding to the corner reflector appears in the range spectrum, and the speed-range heatmap shows that the corner reflector has a velocity of 0.

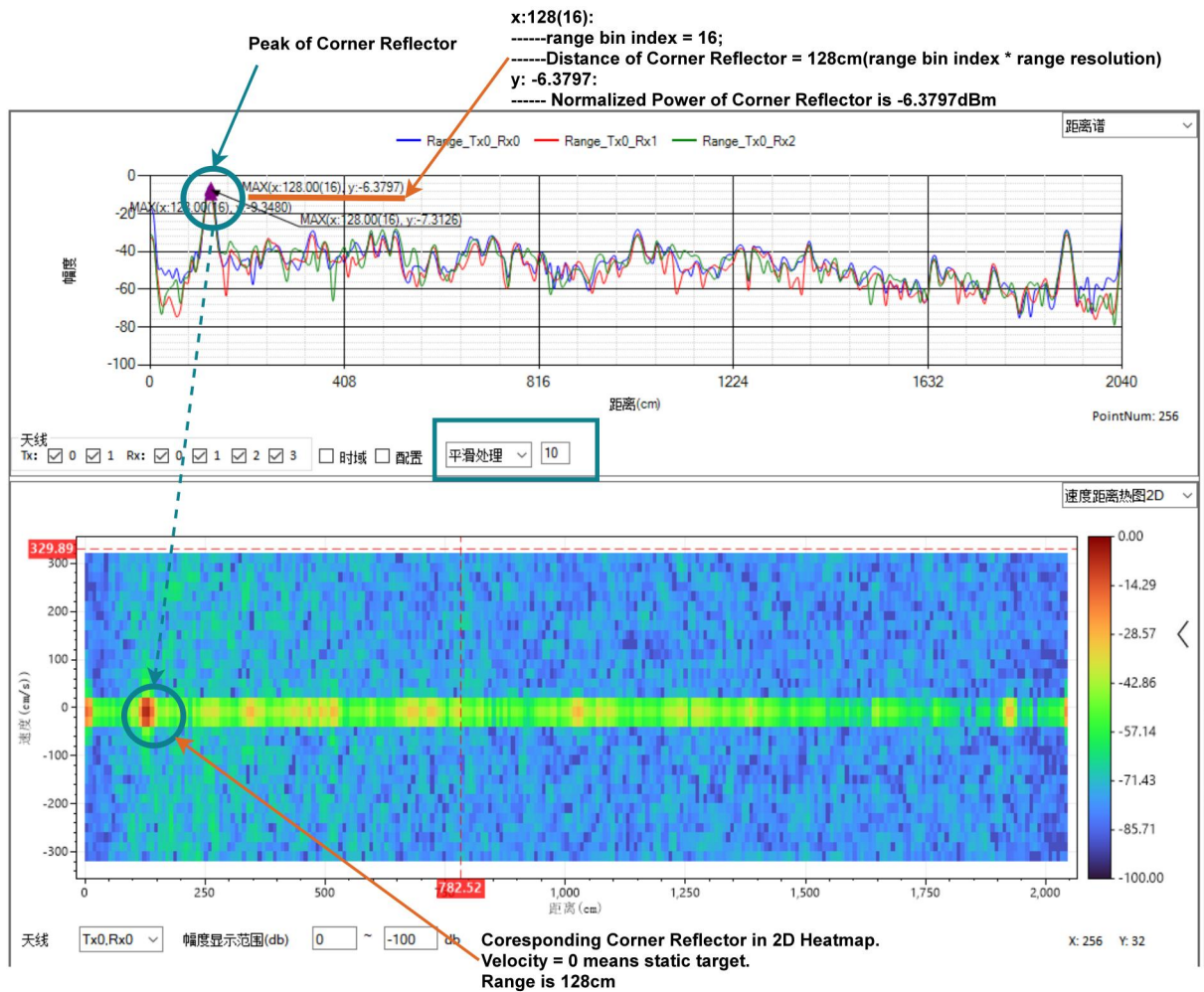


Figure 4-6 Observing a Stationary Target Using Smoothing

- IQ data plot: The IQ data plot maps the 0-velocity component of a specified range bin (and 1 adjacent bin) onto the complex plane. Assuming that the IQ data is a + bj, the functions in the "IQ Data Plot" are as follows:
 - "IQ Phase Plot": Displays $\text{atan2}(b/a)$, in degrees;
 - "IQ Value Plot": Displays the normalized IQ data: $(a+bj)/32768$;
 - "IQ Scatter Plot": Uses the real part as the horizontal coordinate and the imaginary part as the vertical coordinate to plot the position of $(a+bj)/32768$ on the complex plane.

The "IQ Data Plot" is mainly used to analyze the phase stability of a stationary target and is an important phase-analysis method for detecting slowly varying signals such as respiration and heart rate.

4.2 Definitions and Constraints for Radar Performance Evaluation Test Scenarios

Radar performance evaluation requires a stable and reliable test environment, which is critical to obtaining correct performance test results. In an ideal test scenario, there is exactly one intended target within the radar's effective detection range and no objects elsewhere. The radar therefore receives only the echo from that target, while noise at other ranges is generated solely by the radar receive channels.

Because such an ideal scenario generally does not exist, constraints must be imposed on common environments to establish a benchmark test scenario. The test scenario is defined as follows:

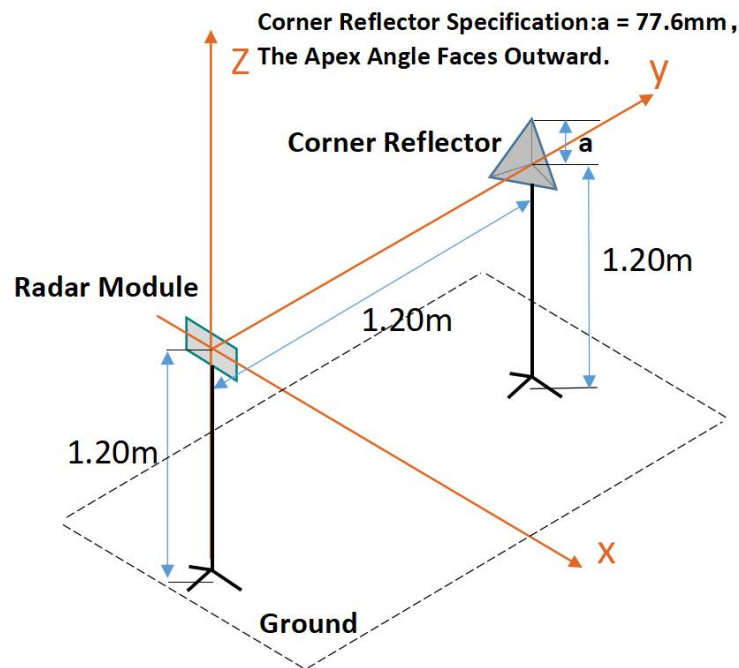


Figure 4-7 Performance Evaluation Environment Requirements

Support both the radar and the corner reflector at a height of 1.2 m using sturdy, stable, nonmetallic stands made of plastic, wood, carbon fiber, or similar materials. Do not allow the radar cable harness to move, as this may cause packet loss or false targets resulting from slight radar vibration.

Define the origin of the radar coordinate system at the center of the receive antenna array, with the x/y/z-axis directions shown in Figure 4-7. The center of the corner reflector lies on the radar boresight (y-axis). Pay particular attention to the following constraints in the test environment:

1. Keep the environment open and completely stationary
 - (1) The radar is sensitive to targets with slight movement or vibration; therefore, no vibrating targets may be present in the environment. Such targets include not only visibly moving objects, but also easily overlooked sources of slight vibration, such as fans, metal objects moved by airflow, metal pipes carrying flowing water, and air-conditioning vents;
 - (2) Keeping the environment open reduces multipath reflections and the evaluation risk caused by possible vibration of stationary objects. Avoid strong metal reflectors or metal ceilings near the test area, and avoid testing in confined spaces. For the test scenario shown in Figure 4-7, keep the area within a 3 m radius of the radar clear and ensure that there are no large metal reflectors beyond the 3 m radius;
 - (3) Avoid testing in windy conditions, as wind may move the radar or corner reflector and interfere with the evaluation results.
2. Test personnel and the test computer should remain behind the radar and as far away from it as possible. Some radar energy is radiated backward, and a person behind the radar may be detected through multipath reflections. Test personnel should therefore remain as far away from the radar as possible; a distance of more than 1 m behind the radar is recommended.
3. Place a corner reflector 1.2 m directly in front of the radar to ensure sufficient target echo energy
 - (1) At 77 GHz, a corner reflector with an RCS of 10 dBsm is recommended. Its dimensions are as follows:

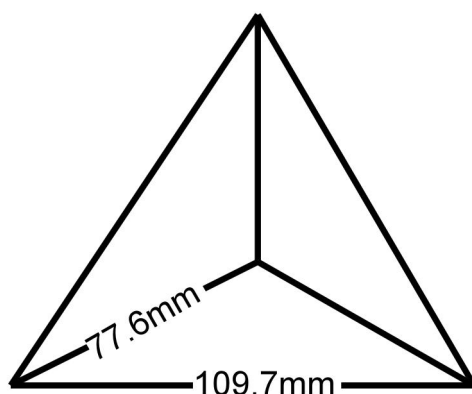


Figure 4-8 Corner Reflector Dimensions

(2) A corner reflector is strongly recommended instead of another object as the test target. A corner reflector can be regarded by the radar as a point target, which eliminates certain other interference factors.

4.3 Noise Floor Performance Evaluation

After completing the operations shown in Figure 4-1, evaluate noise-floor performance by observing the power in the "Range Spectrum." Select "Static Removal" by default and set the "Static Removal" value to the number of Doppler FFT points/4 so that the portion retained by the "Static Removal" function is not affected by stationary-target energy, allowing observation of the statistical characteristics of the noise floor. The "Data View" is shown in Figure 4-9:

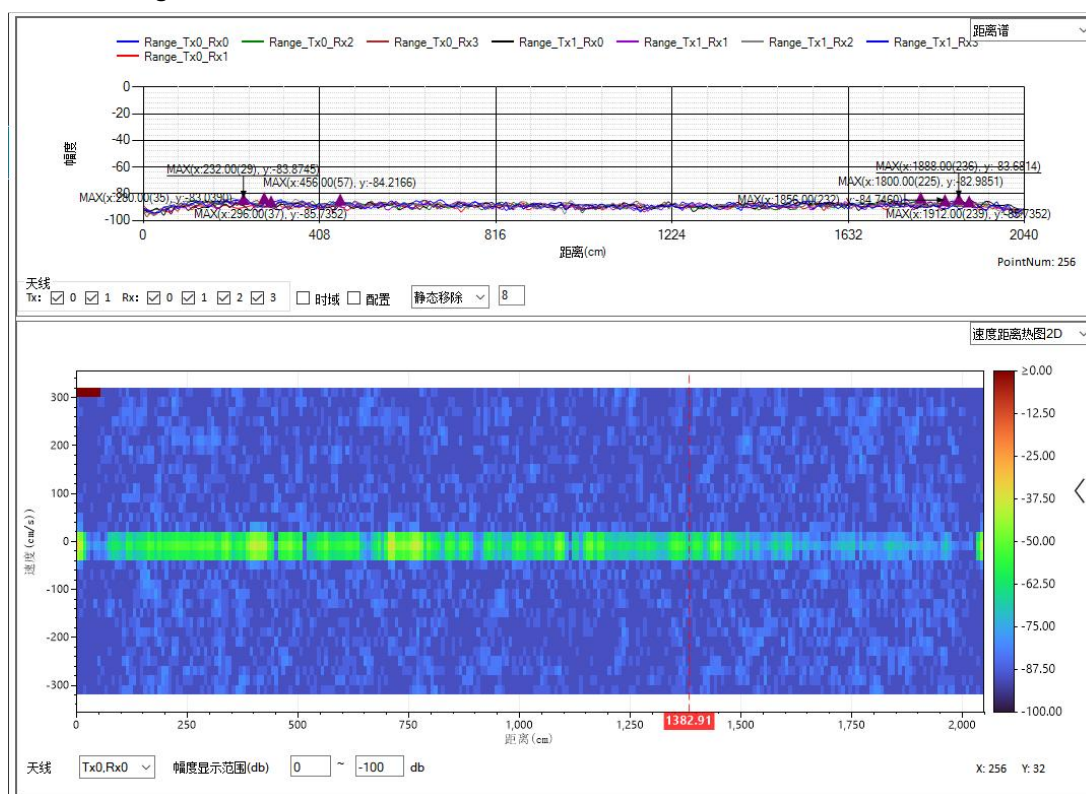


Figure 4-9 Default Page for Noise Floor Performance Evaluation

An ideal noise floor should have the following characteristics:

1. The noise-floor shape should be consistent across all channels;
2. Except at short and long ranges, where it is affected by the high-pass filter, the noise-floor shape should remain essentially flat, with similar noise-floor power at all ranges;
3. The far-range noise-floor power should be between -85 and -95 dB. If it is abnormal, check whether "ETF" -> "Static Configuration" -> "RF Gain Settings," "BB Gain Adjustment," and "Tx Power Configuration" use the default values shown in the figure;

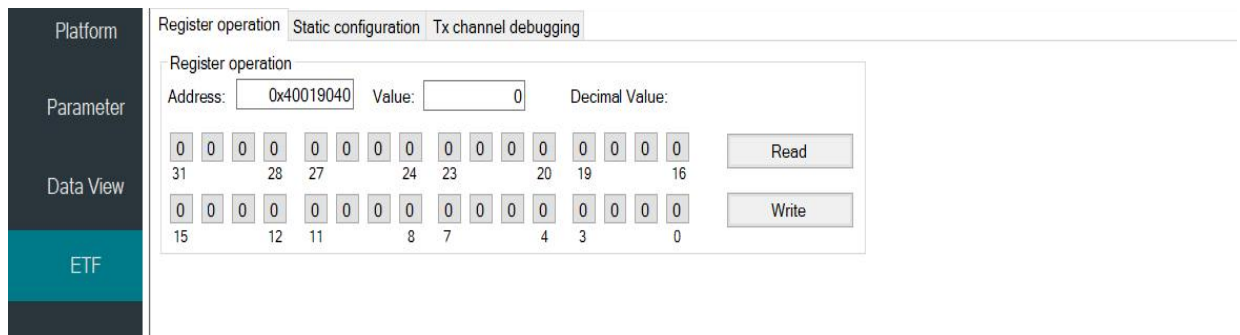


Figure 4-10 ETF Default Configuration

4. No significant large fluctuations should occur in the noise floor during short-term observation;

After confirming in RadarAnalysisTool that there are no obvious noise-floor issues, further evaluation of noise-floor performance requires an environmental soak test lasting at least overnight to determine whether the noise-floor energy in any range bin exhibits abnormal fluctuations.

A typical range spectrum with a normal noise floor is shown below:

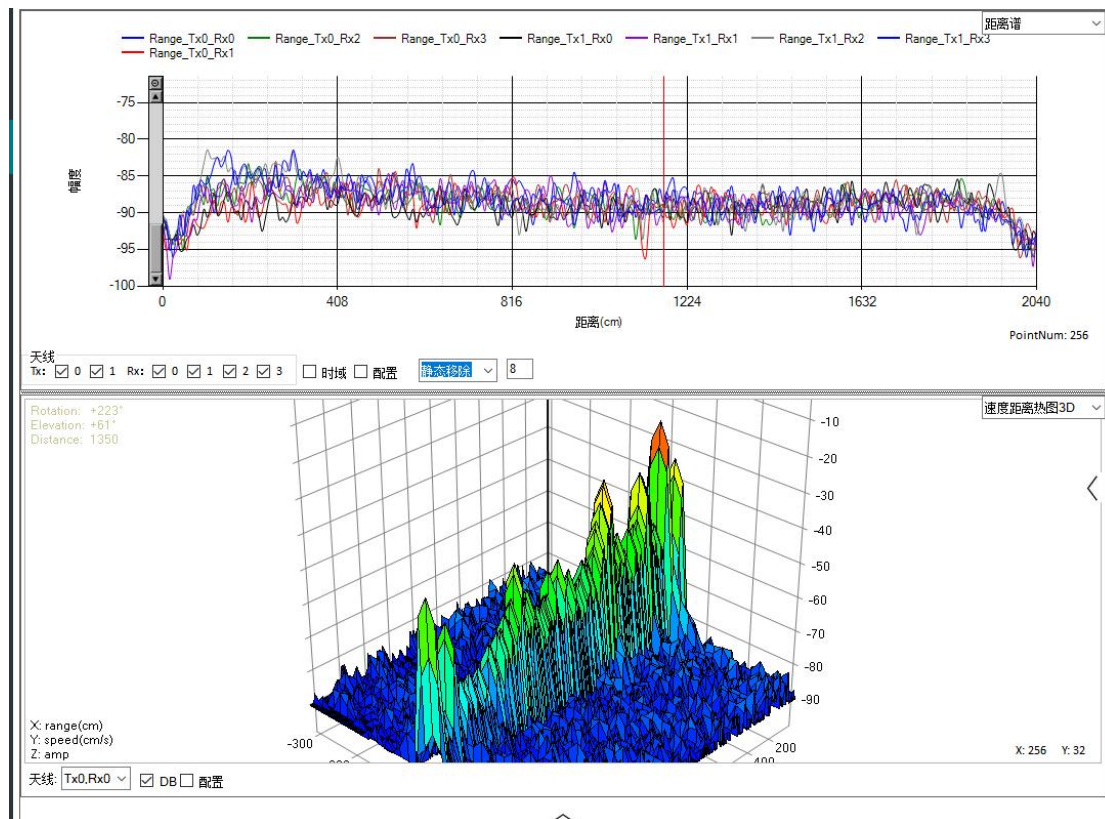


Figure 4-11 Noise Floor Range Spectrum Benchmark

Typical causes of abnormal noise-floor performance include:

1. Short-range Tx-to-Rx interference: **This may cause "humps" in the short-range noise floor of some receive channels.** These "humps" are related to the chip and mechanical structure. The chip itself has slight Tx-to-Rx interference. Adding a radome, PCB, or other structural component may make the "humps" more severe. Developers can assess whether the structural design is reasonable based on changes in the "humps."

The effect of a "hump" is that, at short range, the signal-to-noise ratio for target detection deteriorates on the affected antenna. This issue must be handled separately in subsequent algorithms. Tx-to-Rx interference on the development board is shown below:

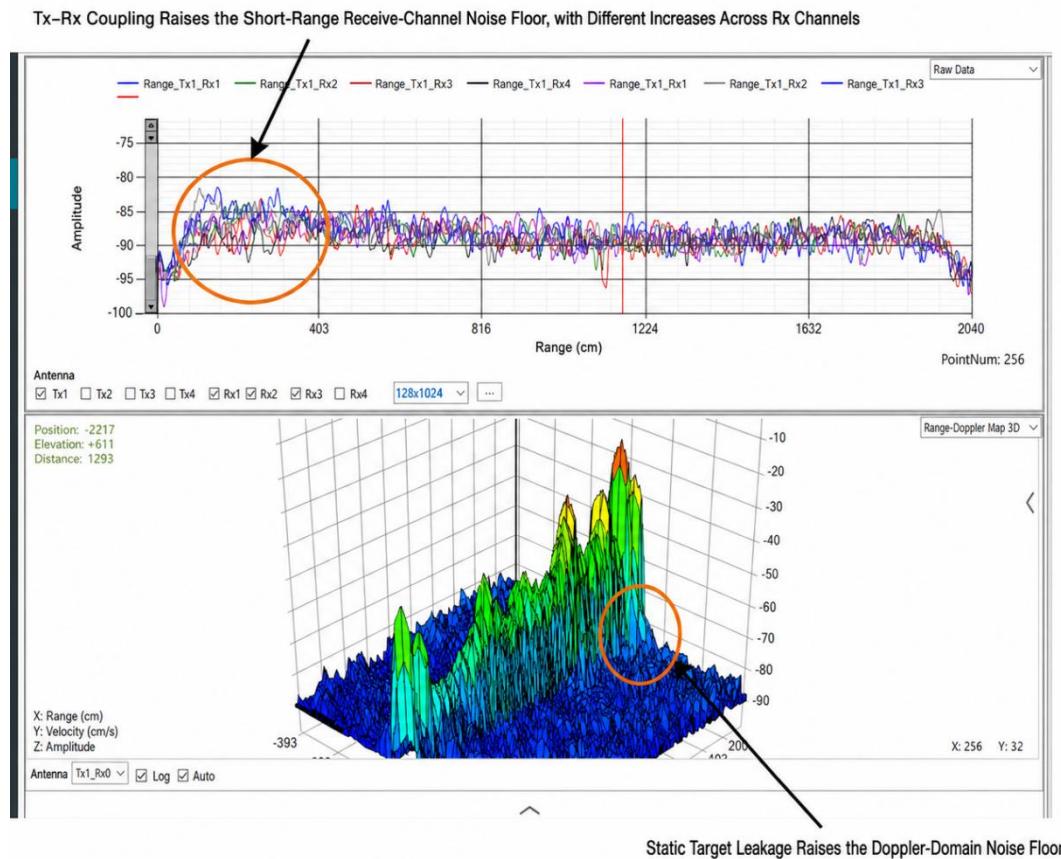


Figure 4-12 Tx-to-Rx Interference

2. Check whether the overall noise floor is flat and whether it rises abnormally at certain ranges. Radar chips undergo rigorous calibration and screening at the factory, so an uneven noise floor generally does not occur. Abnormal rises at specific positions in the noise floor are commonly caused by the power supply or the test environment.

The position of power-supply interference does not change with the environment. Try replacing the DC-DC converter or powering the radar directly with an LDO to resolve the issue. Environment-induced interference changes when the environment changes; in this case, use a different test environment.

4.4 Corner Reflector Detection Performance Evaluation

The purpose of corner-reflector detection performance evaluation is to assess the amplitude and phase stability of each radar channel for a known point target. Evaluate corner-reflector detection performance by switching to the "Range Spectrum" view. Focus on the following indicators to determine whether performance is normal:

1. Target power consistency: On the "Range Spectrum" page, enable "Smoothing" and set the number of smoothing Frames to 10. Check the following in the range spectrum:

- (1) Whether the displayed position of the corner reflector matches its actual position and whether the detected corner-reflector power is consistent across channels. In general, the power difference between any two channels should be less than 5 dB;
- (2) Check whether the absolute power meets the requirement: Corner-reflector power should be approximately -20 dB.

A common issue is inconsistent gain across channels. If certain antennas are obstructed by the mechanical structure, large gain differences may occur and must be handled separately in subsequent algorithms;

2. Target phase stability: Select "IQ Scatter Plot" in the "IQ Data Plot," select the range bin containing the corner reflector, and observe whether the corner reflector's position drifts on the complex plane. During short-term observation, the corner reflector should exhibit only slight jitter around a fixed position on the complex plane;

Typical issues in corner-reflector detection performance evaluation:

1. Power saturation. Using a target with a higher RCS for performance evaluation may saturate the receive channel or baseband processing. When the receive channel is saturated, the spectrum becomes unpredictable. In this case, reduce "RF Gain Settings," increase the HPF cutoff frequency, and reduce "BB Gain Adjustment" from the ETF default configuration shown in Figure 4-10. As a simple criterion for power saturation, the power value should never exceed -15 dB under any condition; otherwise, saturation may occur. A saturated waveform is shown below. This saturation condition can be reproduced by placing a metal plate 2-3 cm in front of the radar and adjusting its angle.

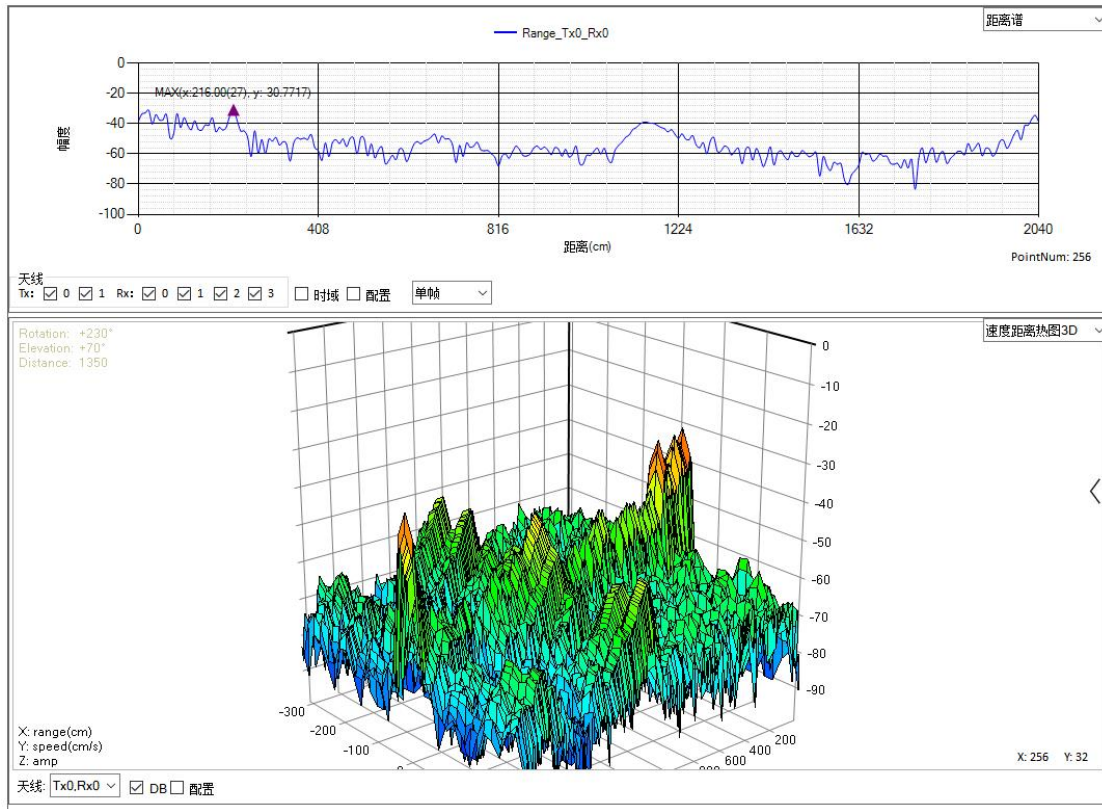


Figure 4-13 Typical Saturated Waveform

There is currently no method to determine whether saturation is occurring, so developers must design an appropriate gain setting based on their own worst-case conditions. In general, the default configuration covers most scenarios.

2. Phase drift. For a corner reflector, which can be regarded as a point target with a high signal-to-noise ratio, the phase of the radar echo from the corner reflector should ideally follow a Gaussian distribution. However, because of the temperature characteristics of the crystal oscillator, the corner-reflector phase may drift as the ambient temperature changes. During performance evaluation, focus on whether phase-noise variance and temperature-induced phase drift affect the algorithm application. In addition, a relative displacement of several hundred micrometers between the radar and the corner reflector is reflected in the phase. Therefore, insufficiently secure mounting of the corner reflector or radar, or pulling by a power cable, can also cause phase drift.

4.5 Sensitivity Evaluation

The evaluations above are all performed on the radar's raw DataCube output. After the waveform configuration is finalized, developers can perform range-extension tests at different angles using the intended waveform configuration and scenario, and qualitatively evaluate target-detection sensitivity by checking whether the target can be observed in the "Range Spectrum."

To prevent stationary targets from interfering with sensitivity evaluation, select "Static Removal" and set the static-removal value to 1.

Detection sensitivity depends heavily on waveform design and subsequent processing algorithms. Developers are therefore advised to save the DataCube during range-extension testing and process the data offline on a PC to evaluate radar sensitivity accurately.

4.6 Algorithm Development Using DataCube

The DataCube reported by RadarAnalysisTool is uncalibrated. If this DataCube is used for angle estimation, the resulting angle may be incorrect. Developers who need to use the DataCube for subsequent algorithm development on a PC can calibrate the DataCube IQ data in either of the following ways:

1. Calibrate the DataCube using the developer's own calibration method;
2. Calibrate the DataCube using the recommended calibration method, which applies only to RS6130 and RS6240 chips.

4.6.1 Calibrating the DataCube

This section introduces a simple calibration method that allows developers to calibrate the DataCube output using default parameters. The core principle is to obtain the IQ data of all DataCube channels and multiply the IQ data of each channel by its calibration coefficient to obtain calibrated IQ data. The pseudocode is as follows:

```

for range_idx = 1 : rfft_num      // loop for all range index
    for dop_idx = 1 : dfft_num    // loop for all dopper index
        for ant_idx = 1 : N
            // multiply by calib_data to calibration IQ data in data cube
            data_cube_calibed(range_idx, dop_idx, ant_idx) =
                data_cube(range_idx, dop_idx, ant_idx) * calib_data(ant_idx);
        end
    end
end
// use data_cube_calibed for later signal process

```

Because both RS6130 and RS6240 use AiP antennas, their calibration parameters are essentially stable. The calibration coefficients for RS6130 and RS6240 (calib_data in the pseudocode) are:

```

calib_data for RS6130: [0.3877+0.4028i,    0.4511+0.5i,   -0.5347-0.3667i];
calib_data for RS6240: [
    0.6566,        0.6956-0.2658i,   -0.9922+0.2330i,   0.6785-0.2128i,   //Rx0~Rx3
    0.6568-0.000i, 0.7045-0.2572i,   -0.9794+0.2218i,   0.6776-0.2127i   //Rx4~Rx7
];

```

4.6.2 Obtaining Calibration Values from the Radar Chip Flash

The calibration values in the radar chip flash are used for antenna calibration. Obtain them as follows:

After connecting the Device in RadarAnalysisTool, enter the command "mmwc ant calib_get" in the "Data Sending Area" at the lower right of the tool interface (note that spaces are required between mmwc, ant, and calib_get), and then click "Start Sending." The calibration values are displayed in the "Data Receiving Area." As shown in Figure 4-14, the values in the orange box are the calibration values stored in the chip flash. For RS6130, the first three calibration values correspond to antennas Rx0-Rx2. For RS6240, the eight calibration values correspond to antennas Rx0-Rx7. The default calibration parameters obtained at this point are fixed-point values: Q11 for RS6130 and Q15 for RS6240. To obtain floating-point values, divide the data by 2^{15} (or 2^{11} for RS6130).

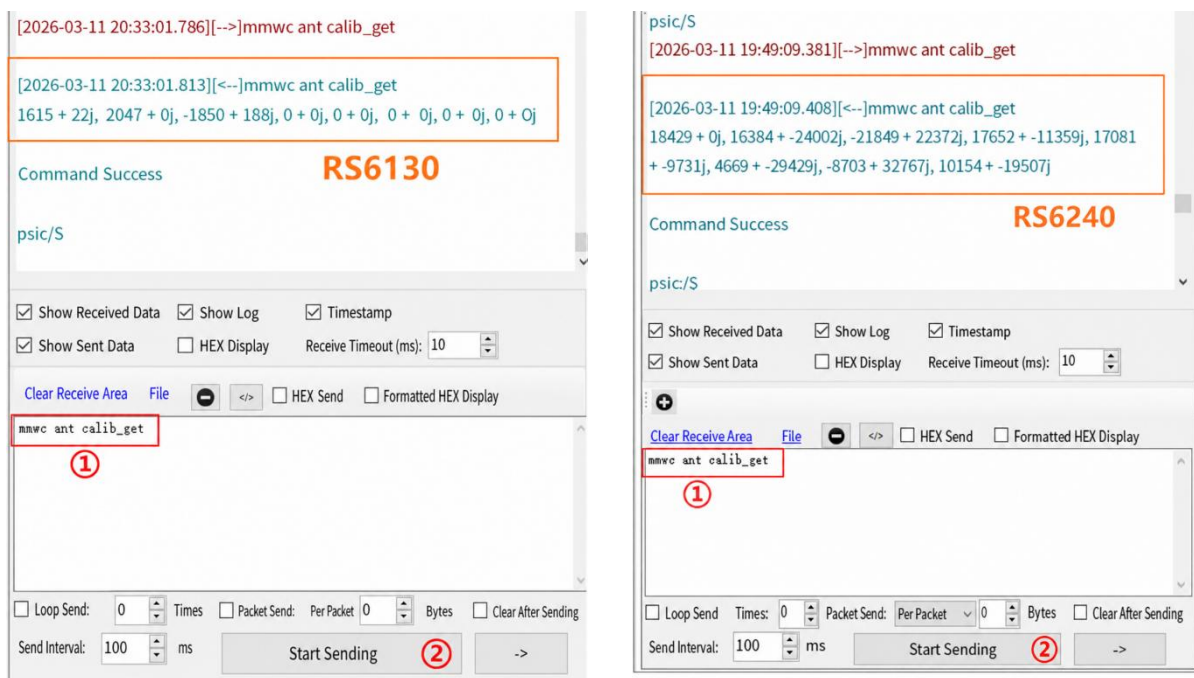


Figure 4-14 Obtaining Calibration Values from the Radar Chip Flash

5 Point Cloud Quality Evaluation Methods

During the development stage, the SDK provides source code for moving-point-cloud and presence-point-cloud algorithms. These algorithms and interfaces are also used by the RadarAnalysisTool firmware. If developers want to use the moving-point-cloud and presence-point-cloud algorithms included in the RadarAnalysisTool firmware as the basis for subsequent development, or use the radar's default point-cloud output as a benchmark during development, they can evaluate the radar point-cloud output.

Point-cloud performance can be evaluated in 3 ways:

1. Test the point-cloud detection limit through range-extension testing;
2. Configure the posture point cloud and perform posture point-cloud testing;
3. Configure different point-cloud filtering levels to test resistance to false targets.

Regardless of whether RadarAnalysisTool is configured for horizontal or vertical placement, the positive directions of the point-cloud output coordinate system are defined as shown below:

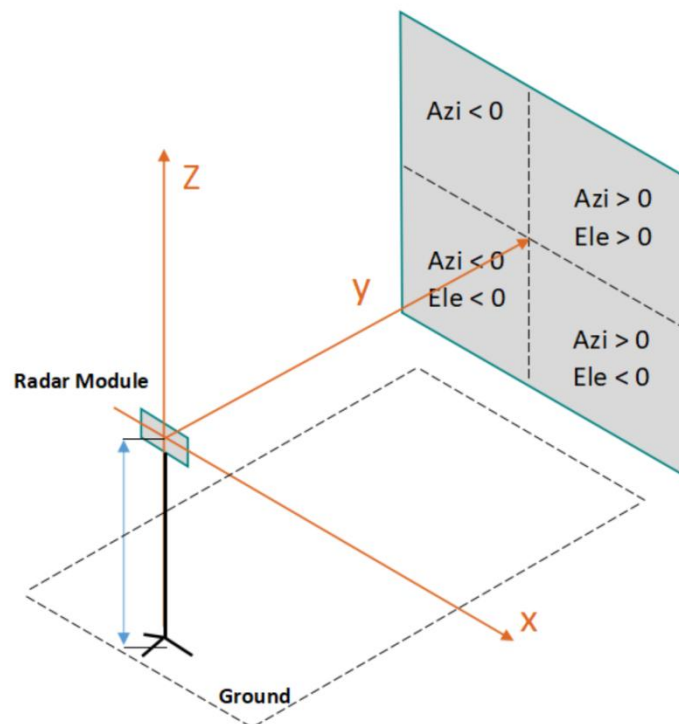


Figure 5-1 Positive-Direction Definition for the Point Cloud

Both RS624x and RS613x support presence-point-cloud output. Right-click in the point-cloud display area and select the option to display the presence point cloud.

5.1 Point Cloud Range-Extension Test

Point-cloud range-extension testing is generally used in outdoor scenarios to evaluate whether the point-cloud output covers the entire intended detection range under the intended waveform configuration. If developers are evaluating the radar for the first time and do not know how to select an appropriate waveform, they can use the waveform configuration shown by clicking "Moving Point Cloud/Tracking Display" on the "Platform Configuration" page.

Developers are advised to install the radar as shown in Figure 5-2 (radar height: 1.2 m; radar boresight parallel to the ground; no obstructions in front of the radar):

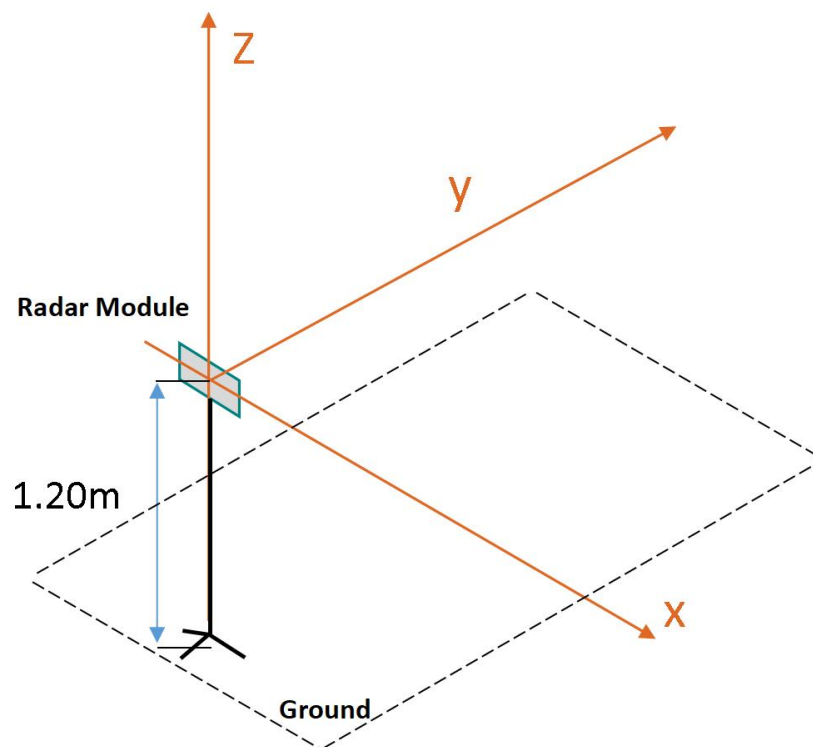


Figure 5-2 Radar Installation Orientation for Range-Extension Testing

A basic range-extension test generally evaluates detection at the edges of the radar FOV (± 60 degrees) and at its center (0 degrees). The horizontal radar field of view (FOV) is indicated by the dashed lines in Figure 5-3.

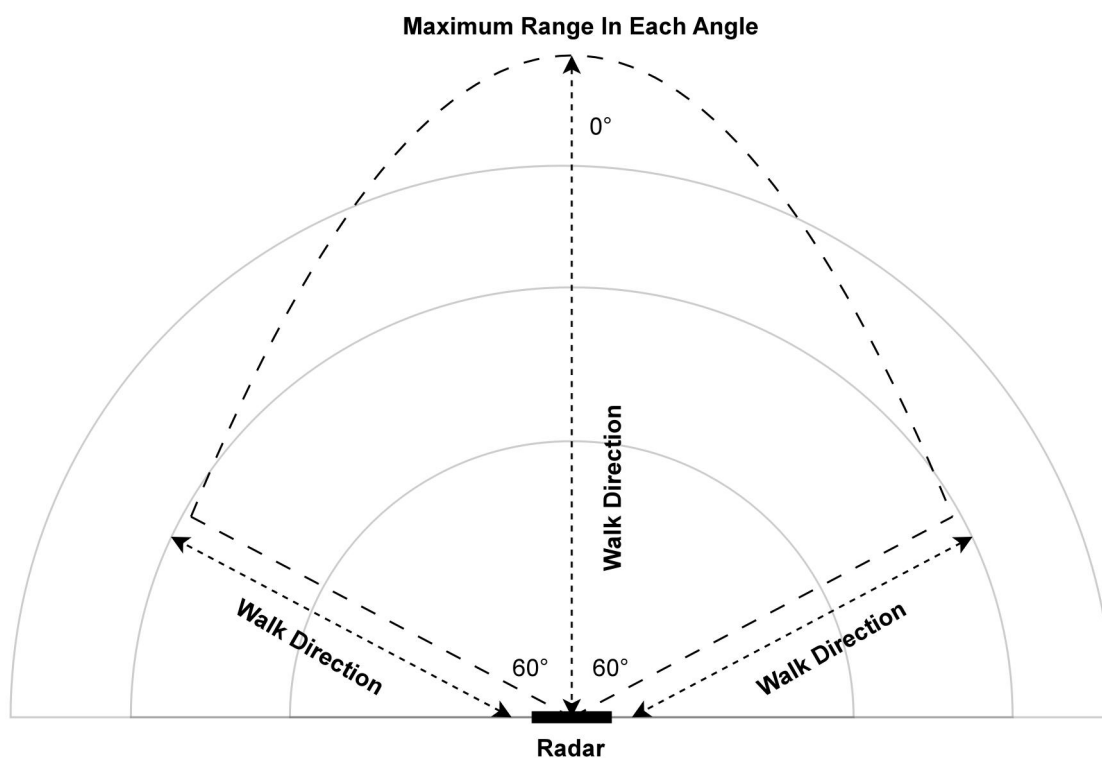


Figure 5-3 Radar Detection Range

If a turntable is available, place the radar on it and rotate the radar to obtain accurate angles and repeatable walking trajectories, thereby ensuring test accuracy.

If the configuration above does not achieve the intended detection range, adjust the following parameters in RadarAnalysisTool to extend the detection range:

The screenshot displays the RadarAnalysisTool configuration interface, organized into several sections:

- Radar Parameter Configuration:** Includes Antenna Mode (1T3R), Start freq (58 GHz), Max Range (20.48 m), Range resolution (8 cm), Max Velocity (3.2 m/s), and Velocity resolution (20 cm/s).
- Frame Structure Configuration:** Includes Frame Structure (2DFFT), Interval Num (32), Frame Num (0), Interval Chirp N (1), Interval Period (397 us), Frame Period (1193 ms), RANGE FFT EN (ON), and DOP FFT EN (ON).
- Configure Calculation Results:** Displays Valid Max Range (20.48 m), Valid Range Resol (8.0 cm), Range Unit (8.0 cm), Radar Tx Duration (12.3ms), Valid Max Velocity (3.2 m/s), Valid Velocity Resol (20.0 cm/s), Doppler (20.0 cm/s), Range FFT (256), and Doppler FFT (32).
- Chirp Parameter:** Includes End Freq (60069MHz), Slope (73MHz/us), Valid Strat Freq (58157MHz), Sample Time (25us), Ramp Time (28us), Sweep BW (2069MHz), Valid BW (1875MHz), and Effective ADC sampling points (256).
- PointCloud Report Data Configuration:** Includes Moving PointCloud Config (SNR(dB) 17, CFAR: SNR Thresho, StaticObjectRemove checked) and Presence PointCloud Config (SNR(dB) 15, FrmDecRate: 4, EnablePresencePointCloud checked).
- MMW Frame Diagram:** A timeline diagram showing the sequence of chirps (chirp 0-N, Idle, chirp 0-N, Idle, chirp 0-N), intervals (Interval 0, Interval 1, Interval N), and frame components (Frame, Data Proc, Frame Idle, PowerSave).
- Clutter Elimination:** Coefficient (0) and a toggle switch (OFF).
- StartOption:** SendConfigAtStart checked, with Start and Stop buttons.
- Radar Data Report Configuration:** Data Type set to Point Cloud, with a Get Config button.

Figure 5-4 Configurations Related to Detection Range Adjustment

1. Reduce "SNR (dB)," which increases the number of clutter points;
2. Increase the number of range FFT points, the number of Doppler FFT points, or the "Interval Chirp N" value. This may increase power consumption but significantly improves the target signal-to-noise ratio. Each doubling of the number of range FFT points, the number of Doppler FFT points, or the number of chirps increases the maximum detection range to approximately 1.2 times its previous value. For example, if the maximum detection range is 10 m with 256 range FFT points, 16 Doppler FFT points, and chirp=1, then with 256 range FFT points, 32 Doppler FFT points, and "Interval Chirp N"=2, the theoretical maximum detection range becomes $10 \times 1.2 \times 1.2 = 14.4$ m. For information on configuring the numbers of range and Doppler FFT points, see the "Evaluation Using RadarAnalysisTool" section.
3. For recommended waveform configurations, see Section 3.1.

5.2 Posture Point Cloud Application Test

Posture point-cloud applications require dense, highly cohesive, high-Frame-rate point clouds. By configuring a point cloud with high range resolution and high velocity resolution in RadarAnalysisTool and combining it with multi-Frame accumulation display and other methods, developers can evaluate whether the default point-cloud output meets the requirements of posture point-cloud applications.

On the "Platform Configuration" page at the left of RadarAnalysisTool, click "Posture Point Cloud Display." RadarAnalysisTool uses a script to configure the posture point cloud and switches to the recommended display. A typical posture point-cloud application configuration is shown below:

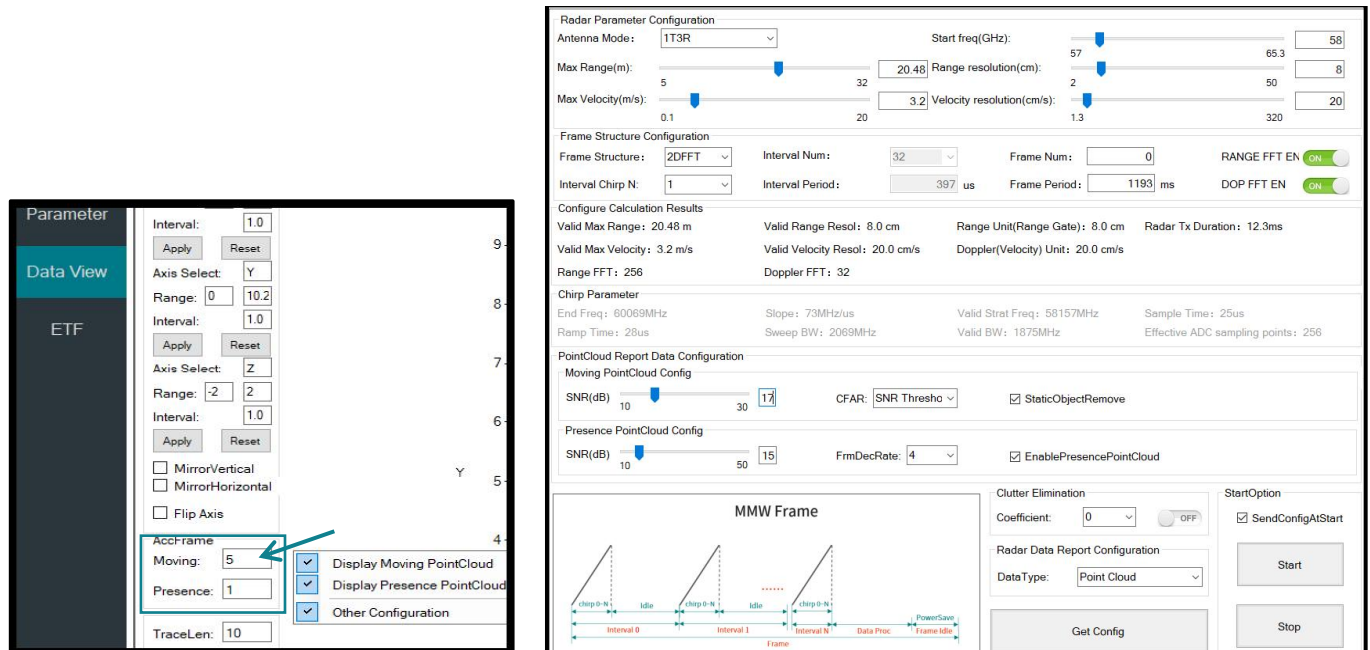


Figure 5-5 Typical Posture Point Cloud Configuration

For better display results, enable accumulated Frames through "Other Configuration" in Data View. Developers can test the posture point cloud using typical movements such as waving, kicking, turning, squatting, and jumping, and qualitatively evaluate whether it meets application requirements. The figure below shows the point-cloud result for one Frame while a person is waving in the posture point-cloud application demo, with orange lines indicating the person's instantaneous posture.

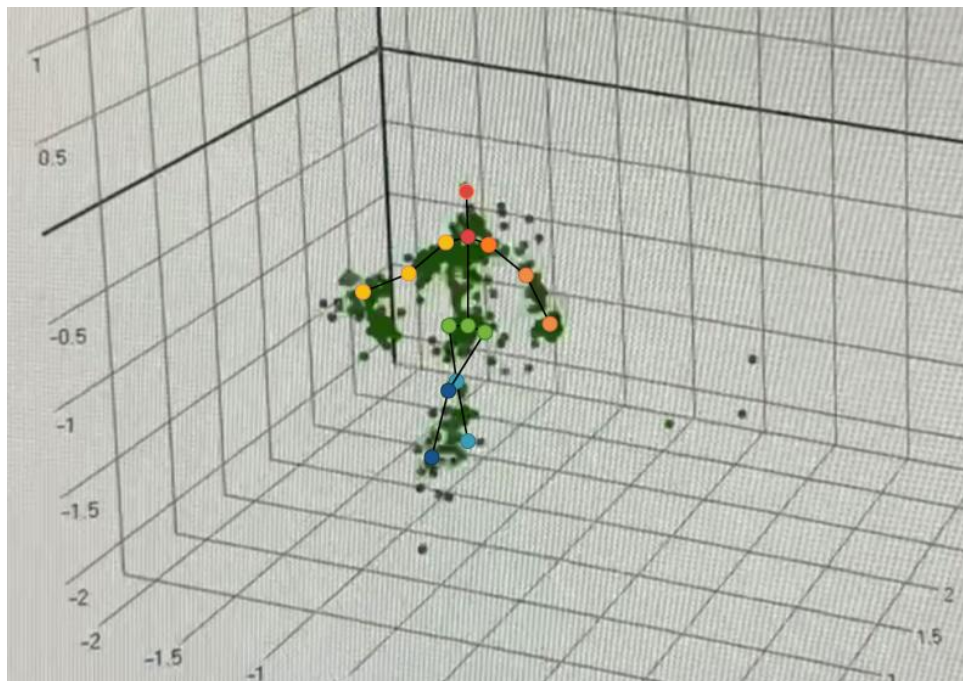


Figure 5-6 Typical Posture Point Cloud Result

For subsequent offline point-cloud processing, use the RadarAnalysisTool saving function to save point-cloud information. Each point includes 3D Cartesian coordinates, Doppler velocity, signal-to-noise ratio, and other information. For instructions on using the RadarAnalysisTool saving function, see the ['RS6x_7x_RadarAnalysisTool_User_Manual'](#).

If posture point-cloud performance does not meet expectations, try the methods in Section 5.1 to improve the signal-to-noise ratio, or use a higher CFAR threshold to reduce clutter points and emphasize the posture point cloud. The RadarAnalysisTool project cannot fully configure the algorithm. For better results, see and try the Posture project in the ['RS6x_7x mmWave_Radar_Sample_Reference_Manual'](#).

5.3 Moving Point Cloud Performance Configuration

Different CFAR algorithms have a significant effect on point-cloud performance. RadarAnalysisTool provides two CFAR algorithms: CA-CFAR and "SNR Threshold." The point-cloud output characteristics of each CFAR algorithm are shown in Table 5-1.

Table 5-1 CFAR Algorithm Selection

	Point Cloud Density	Number of False Targets	Detection Capability	Recommended Environment	Usage Restrictions
CA-CFAR	Low	Few	Poor	Mobile platforms; indoor environments;	-
SNR Threshold	High	Many	Good	Sparsely distributed outdoor targets; indoor environments (weak multipath effects); posture point clouds	Outputs up to 512 point-cloud points; the point-cloud limit is easily reached on mobile platforms

Clear "Stationary Target Removal" to prevent the radar from reporting targets that are stationary relative to the radar;

Use a higher SNR threshold to filter out more moving point-cloud clutter points.

The SDK supports additional moving-point-cloud configurations. For a deeper understanding of moving-point-cloud performance and additional configurations, see the Sample in mmwave_application and the corresponding '[RS6x_7x mmWave_Radar_Sample_Reference_Manual](#)'.

5.4 Presence Point Cloud Performance Configuration

The presence point cloud supports configuration of the SNR threshold and Frame extraction rate. Note: Do not enable stationary clutter suppression when the presence point cloud is enabled.

A higher SNR threshold reduces false detections in the presence point cloud, while a higher Frame extraction rate allows the presence point cloud to observe targets with smaller motion amplitudes.

The SDK supports additional presence-point-cloud configurations. For a deeper understanding of presence-point-cloud performance and additional configurations, see the Sample in mmwave_application and the corresponding '[RS6x_7x mmWave_Radar_Sample_Reference_Manual](#)'.

6 Q&A

Q: Why does RadarAnalysisTool sometimes fail to connect after I click "Connect Device"?

A: The radar takes approximately 5 s after power-on before it can receive shell commands. Wait 5 s after powering on the radar, then click "Connect Device." If the issue persists, disconnect the USB connection, close RadarAnalysisTool, power-cycle the radar, reopen RadarAnalysisTool, and reconnect.

Q: When communication between the radar and RadarAnalysisTool becomes abnormal, why does pressing the reset button on the development board sometimes fail to restore operation?

A: The reset button on the development board does not provide a POR reset. Therefore, in certain special cases, developers must power-cycle the radar and restart RadarAnalysisTool to reset the entire system.

Q: During radar performance testing, why does the spectrum sometimes appear abnormal, with high clutter in the Doppler dimension?

A: Check whether the corner reflector is too close to the radar or too large. Try placing it farther away or using a smaller corner reflector. Check Table 3-1 to determine whether 2x compression is being used. If so, reduce the DataCube dimensions.

Q: Why does detection performance sometimes seem poor during point-cloud range-extension testing?

A: The radar detects only moving targets, specifically those whose velocity has a sufficient component along the line between the person and the radar. This component is called radial velocity. In simple terms, the radar cannot detect a target whose absolute radial velocity is less than the velocity resolution. If a person moves tangentially to a circle centered on the radar, with the distance from the person to the radar as the radius, the radar will not detect the person.

Q: Why does the radar falsely detect stationary targets?

A: Windowing in the Doppler FFT causes the energy of stationary targets to leak into adjacent Doppler bins, as shown in Figure 4-12. During hardware CFAR detection, 1 Doppler bin on each side of a stationary target is excluded from detection. However, if the target energy is too high, leakage into more distant bins may exceed the CFAR threshold, causing CFAR to falsely detect the stationary target. Adjust the CFAR threshold or perform a secondary SNR check in software to eliminate these false detections from stationary targets.

Q: Why does the radar generate many false targets in some environments?

A: In confined environments containing many metal objects, the radar may generate many false targets. These false targets primarily arise from two sources:

1. Targets detected through multiple propagation paths. Typical multipath interference is shown below:

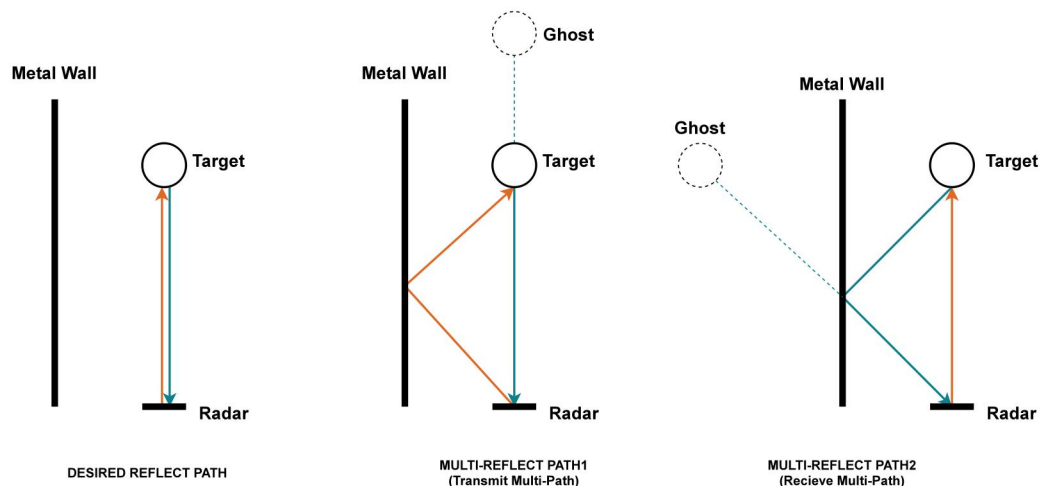
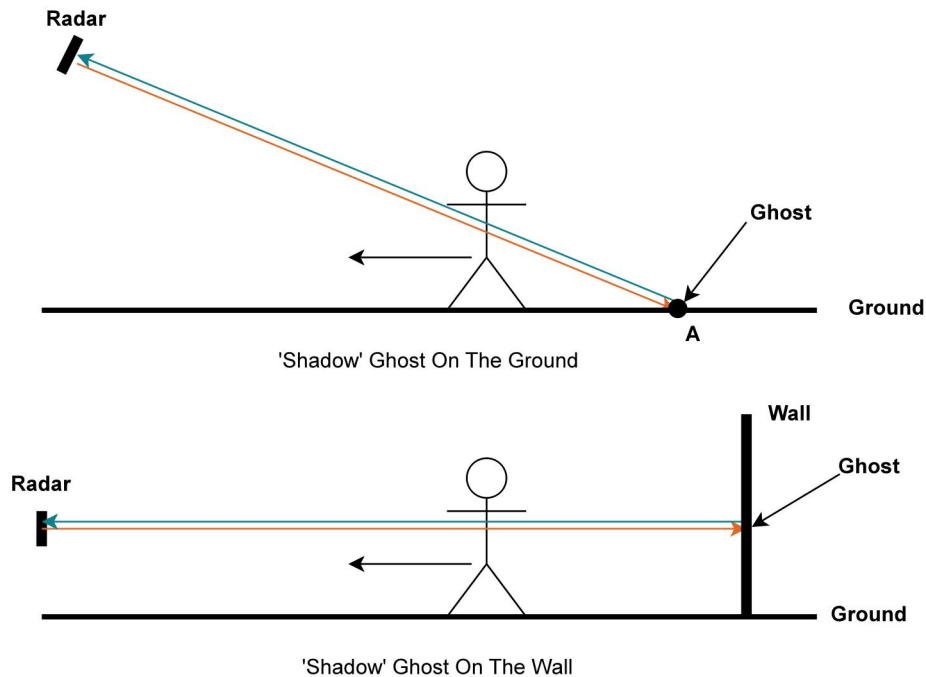


Figure 6-1 Multipath Interference

2. A moving target obscures a stationary target behind it, but the occlusion is not stable. The stationary target's energy is "amplitude-modulated" by the person's motion frequency and enters the velocity dimension, causing the stationary target to be identified as a moving target. If the radar is compared to a light source, these stationary targets are detected as the person's "shadows" from the radar's perspective.

**Figure 6-2 "Shadow" False Target**

Q: Why does RadarAnalysisTool sometimes report a plotting exception during point-cloud performance evaluation?

A: Because of PC performance limitations, when a large number of point-cloud points are reported for an extended period, the plotting control's buffer overflows and the control reports a "Plotting Exception." Place the radar on a stable base to avoid generating a large number of point-cloud points for an extended period, or use a higher-performance PC.

Q: Why does RadarAnalysisTool sometimes report "Incomplete Data Received"?

A1: The clock frequency is high when the radar reports a large amount of data, imposing signal-integrity requirements on the connector. When a low-quality USB cable is used, touching the USB cable harness or Type-C connector may cause signal-integrity issues. Place the radar on a stable base and avoid touching the radar cable harness.

A2: With SPI transmission, the data-transfer duration depends on PC performance and task scheduling and is therefore nondeterministic. As a result, "Incomplete Data Received" may still appear during data transmission even at the minimum Frame rate allowed by RadarAnalysisTool. Manually increase the Frame period or use a higher-performance PC to prevent data errors.

Appendix A:

Analyzing the RD Spectrum of a Saved DataCube (.CSV) File

This section uses an example to show how to obtain analyzable RD spectra (2D speed-range heatmap and 3D speed-range heatmap) from raw IQ data in a .csv file saved by RadarAnalysisTool, providing developers with a more detailed understanding of RadarAnalysisTool output and radar performance evaluation.

For example, in RS6240 multi-antenna 2T4R mode (using 2 transmit antennas and 4 receive antennas), with a 2DFFT Frame structure, 256 range FFT points, and 16 Doppler FFT points, the raw .csv file stores data in the following format:

FrameIdx	IntervalIdx	bin0	bin1	bin2	bin3	...	...	...	bin254	bin255
0	0	1	-1	1	0	-1	-1	1	-2	2
0	0	-1	2	-1	1	-2	6	-10	0	1
0	1	3	0	-1	1	0	1	-5	1	-1
0	1	1	-2	1	-1	1	-3	3	0	0
0	2	-2	1	-1	2	-2	1	3	-2	2
0	2	-1	1	-1	1	0	-1	3	-1	1
0	...	14	-18	17	-10	7	-7	10	-6	-2
0	...	-14	7	7	-18	18	-3	-12	11	0
0	8	-273	261	-168	302	-406	334	-133	69	7
0	8	-485	153	-11	268	-430	375	-151	-93	297
0	...	-16	12	-4	-6	10	-11	9	1	-4
0	...	-3	13	-19	9	-3	3	3	-7	6
0	14	-1	-1	2	1	-1	2	-5	0	1
0	14	-1	1	-1	0	0	1	2	0	0
0	15	2	1	-1	-1	4	-2	5	2	-2
0	15	2	-1	1	0	1	-4	5	0	-1
0	0	0	0	0	0	-2	1	1	0	2
0	0	-1	1	0	0	0	3	-5	2	3
0	1	-1	2	-1	2	0	1	-2	1	0
0	1	1	-1	1	-1	0	-1	3	-4	0
0	2	-2	1	1	0	1	-3	4	-1	-4
0	2	-1	1	-1	1	0	-1	3	-1	1
0	...	12	-116	112	-44	-59	8	81	73	-131
0	...	-85	-33	-65	17	93	-108	255	-291	-14
0	8	1309	-938	441	-843	1026	-711	66	176	5
0	8	1518	-485	11	-351	23	100	162	-71	257
0	...	1	-1	2	-1	1	-2	2	-3	2
0	...	2	-2	3	-1	0	-3	3	4	-7
0	14	-1	2	-2	1	2	0	-3	6	-7
0	14	-2	3	-3	2	2	-1	1	-1	5
0	15	2	1	-1	-1	4	-2	5	2	-2
0	15	2	-1	1	0	1	-4	5	0	-1
0	0	1	-1	0	1	-1	1	0	5	-3
0	0	1	0	-1	1	2	-2	2	-3	1
0	1	0	0	0	1	0	-1	1	-4	4
0	1	0	0	1	0	0	1	-1	2	-5
0	2	1	-2	2	-1	0	1	-2	2	-1
0	2	-1	0	0	1	0	-1	1	5	...
FrameIdx	IntervalIdx	bin0	bin1	bin2	bin3	...	...	...	bin254	bin255
0	0	1	-1	1	0	-1	-1	1	-2	2
0	0	-1	2	-1	1	-2	6	-10	0	1
0	1	3	0	-1	1	0	1	-5	1	-1
0	1	1	-2	1	-1	1	-3	3	0	0
0	2	-2	1	-1	2	-2	1	3	-2	2
0	2	-1	1	-1	1	0	-1	3	-1	1
0	...	14	-18	17	-10	7	-7	10	-6	-2
0	...	-14	7	7	-18	18	-3	-12	11	0
0	8	-273	261	-168	302	-406	334	-133	69	7
0	8	-485	153	-11	268	-430	375	-151	-93	297
0	...	-16	12	-4	-6	10	-11	9	1	-4
0	...	-3	13	-19	9	-3	3	3	-7	6
0	14	-1	-1	2	1	-1	2	-5	0	1
0	14	-1	1	-1	0	0	1	2	0	0
0	15	2	1	-1	-1	4	-2	5	2	-2
0	15	2	-1	1	0	1	-4	5	0	-1
0	0	0	0	0	0	-2	1	1	0	2
0	0	-1	1	0	0	0	3	-5	2	3
0	1	-1	2	-1	2	0	1	-2	1	0
0	1	1	-1	1	-1	0	-1	3	-4	0
0	2	-2	1	1	0	1	-3	4	-1	-4
0	2	-1	1	-1	1	0	-1	3	-1	1
0	...	12	-116	112	-44	-59	8	81	73	-131
0	...	-85	-33	-65	17	93	-108	255	-291	-14
0	8	1309	-938	441	-843	1026	-711	66	176	5
0	8	1518	-485	11	-351	23	100	162	-71	257
0	...	1	-1	2	-1	1	-2	2	-3	2
0	...	2	-2	3	-1	0	-3	3	4	-7
0	14	-1	2	-2	1	2	0	-3	6	-7
0	14	-2	3	-3	2	2	-1	1	-1	5
0	15	2	1	-1	-1	4	-2	5	2	-2
0	15	2	-1	1	0	1	-4	5	0	-1
0	0	1	-1	0	1	-1	1	0	5	-3
0	0	1	0	-1	1	2	-2	2	-3	1
0	1	0	0	0	1	0	-1	1	-4	4
0	1	0	0	1	0	0	1	-1	2	-5
0	2	1	-2	2	-1	0	1	-2	2	-1
0	2	-1	0	0	1	0	-1	1	5	...

Figure A-1 DataCube Storage Format in a CSV File

To visualize the data, for example using an RD spectrum to observe target position and velocity, first correctly reconstruct the complex data from the saved .csv file. The pseudocode is as follows:

```

% data_cube_csv : the original data in the csv file;
% data_cube : the four-dimensional data(frame,ant,interval ,range) matrix available for subsequent
                processing;
% frame_data_row : the number of rows occupied by each frame of data;
% ant_data_row : the number of rows occupied by each ant of data;
% frame_data_row = ant_data_row * ant_num;
% ant_data_row = 2 * interval_num;

for frame_idx = 1:frame_num
    for ant_idx = 1:ant_num
        % find the starting row index of each antenna within the frame
        start_data_row = 1 + (frame_idx - 1) * frame_data_row + (ant_idx - 1) * ant_data_row;
        for interval_idx = 1:interval_num
            real_data_row = start_data_row + (interval_idx - 1) * 2;
            imag_data_row = real_data_row + 1;
            real_data = data_cube_csv(real_data_row);
            imag_data = data_cube_csv(imag_data_row);
            % the normalization complex data of all range bin
            complex_data = (real_data + 1i * imag_data) ./ 2^15;
            data_cube(frame_idx,ant_idx,interval_idx,:) = complex_data;
        end
    end
end

% use data_cube for later signal process
%% take drawing the RD spectrum of a single - frame TX0RX0 antenna of the RadarAnalysisTool as an
example
data_cube_single_frame_tx0rx0 = data_cube(1,1,:);
data_cube_power_single_frame_tx0rx0 = abs(data_cube_single_frame_tx0rx0).^2;
% drawing 3D RD spectrum
figure(1);
mesh(db(data_cube_power_single_frame_tx0rx0, 'power'));
title("Speed-Range Heatmap 3D")
% drawing 2D RD spectrum
figure(2);
mesh(db(data_cube_power_single_frame_tx0rx0, 'power'),
view(2);
title("Speed-Range Heatmap 2D")

```

The "Speed-Range Heatmap 2D" and "Speed-Range Heatmap 3D" in the RadarAnalysisTool Data View are generated by the example above. The corresponding RadarAnalysisTool views are shown below:

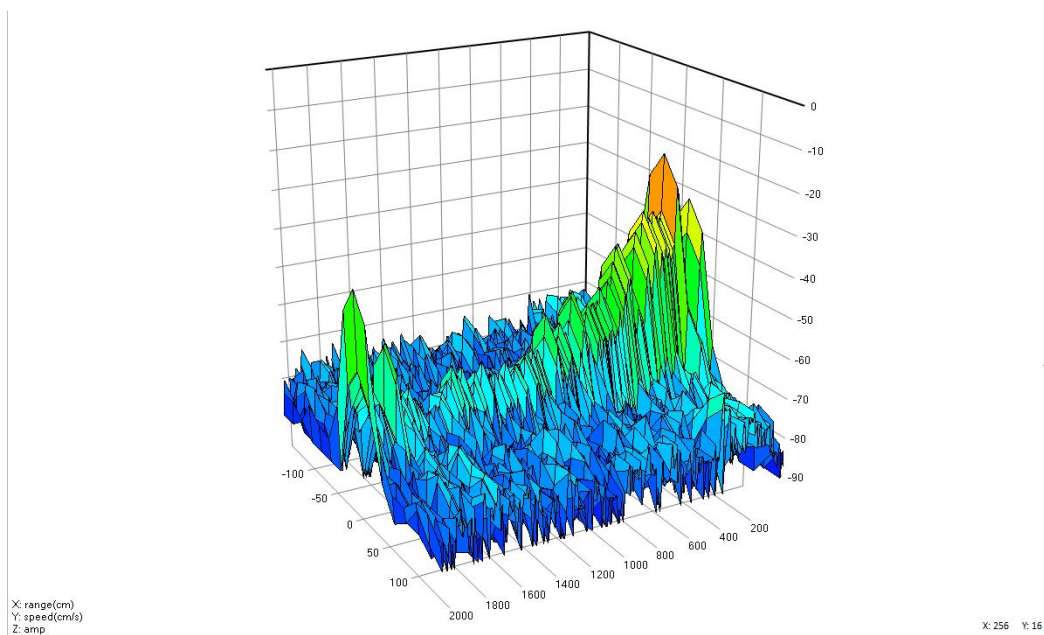


Figure A-2 Speed-Range Heatmap - 3D

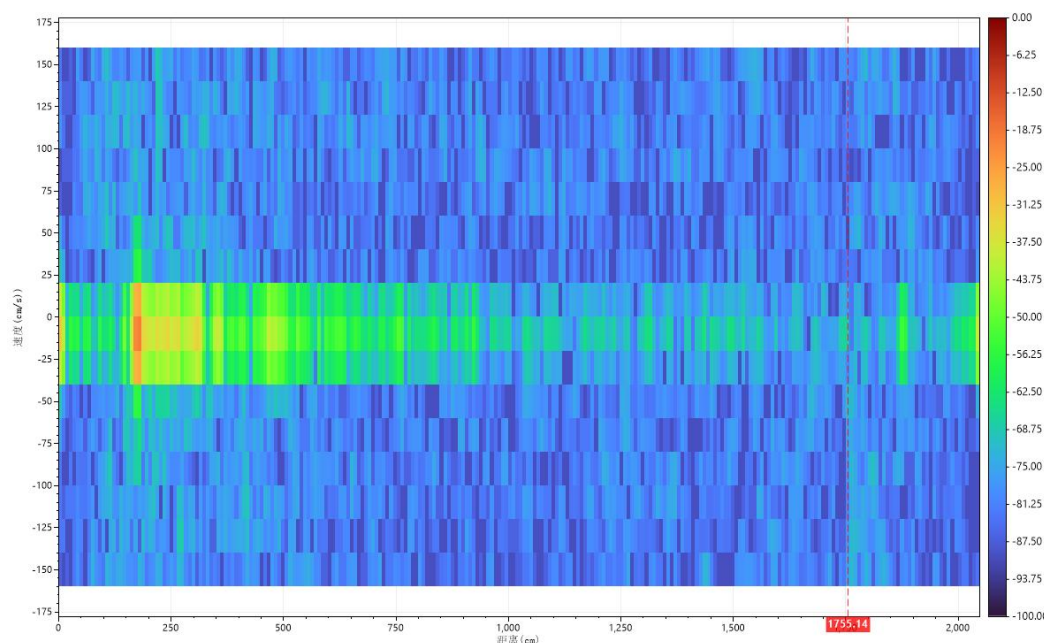


Figure A-3 Speed-Range Heatmap - 2D

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